

# BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

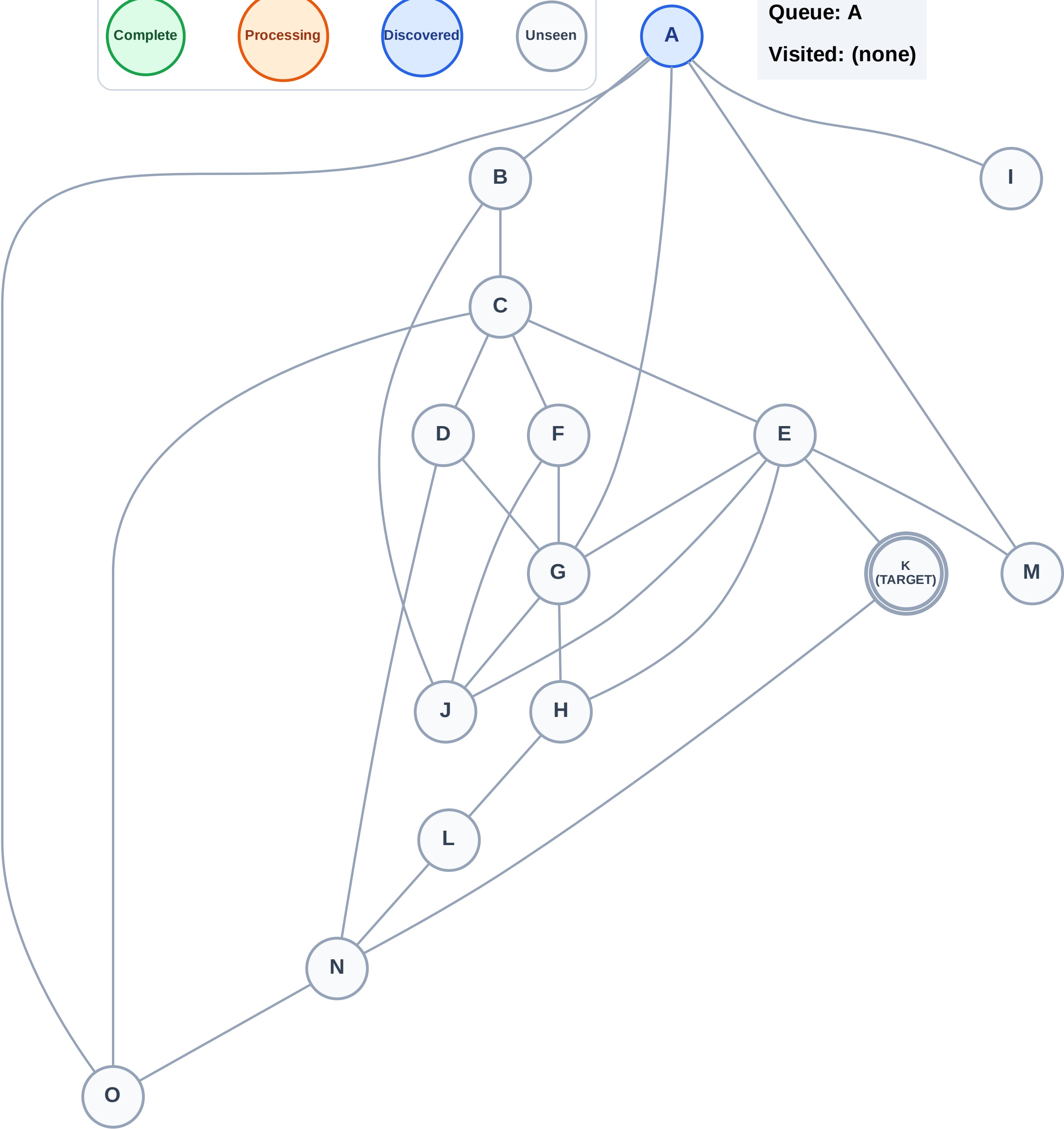
Processing

Discovered

Unseen

Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

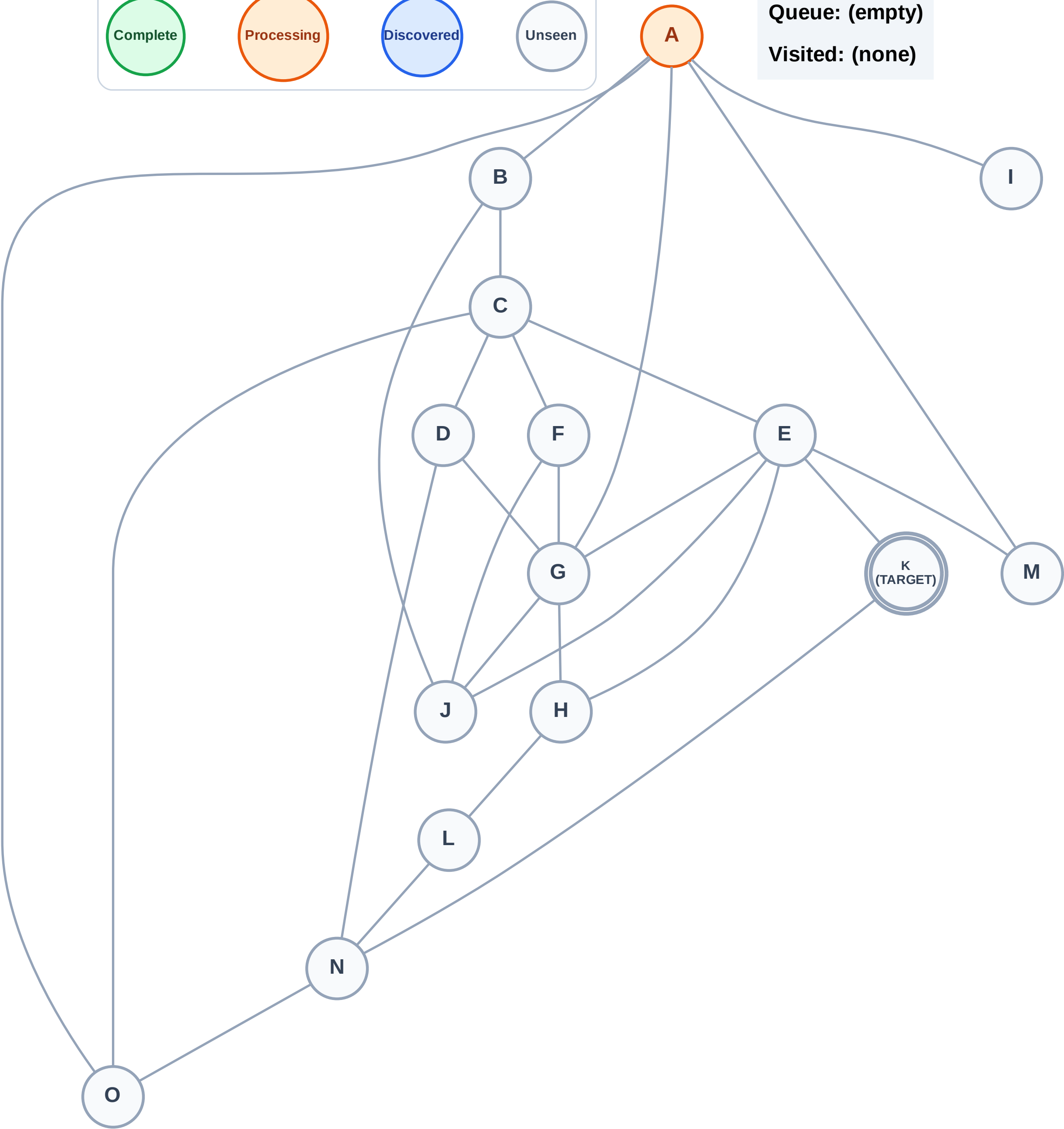
Complete

Processing

Discovered

Unseen

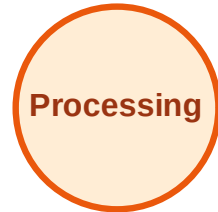
Queue: (empty)  
Visited: (none)



# BFS Traversal

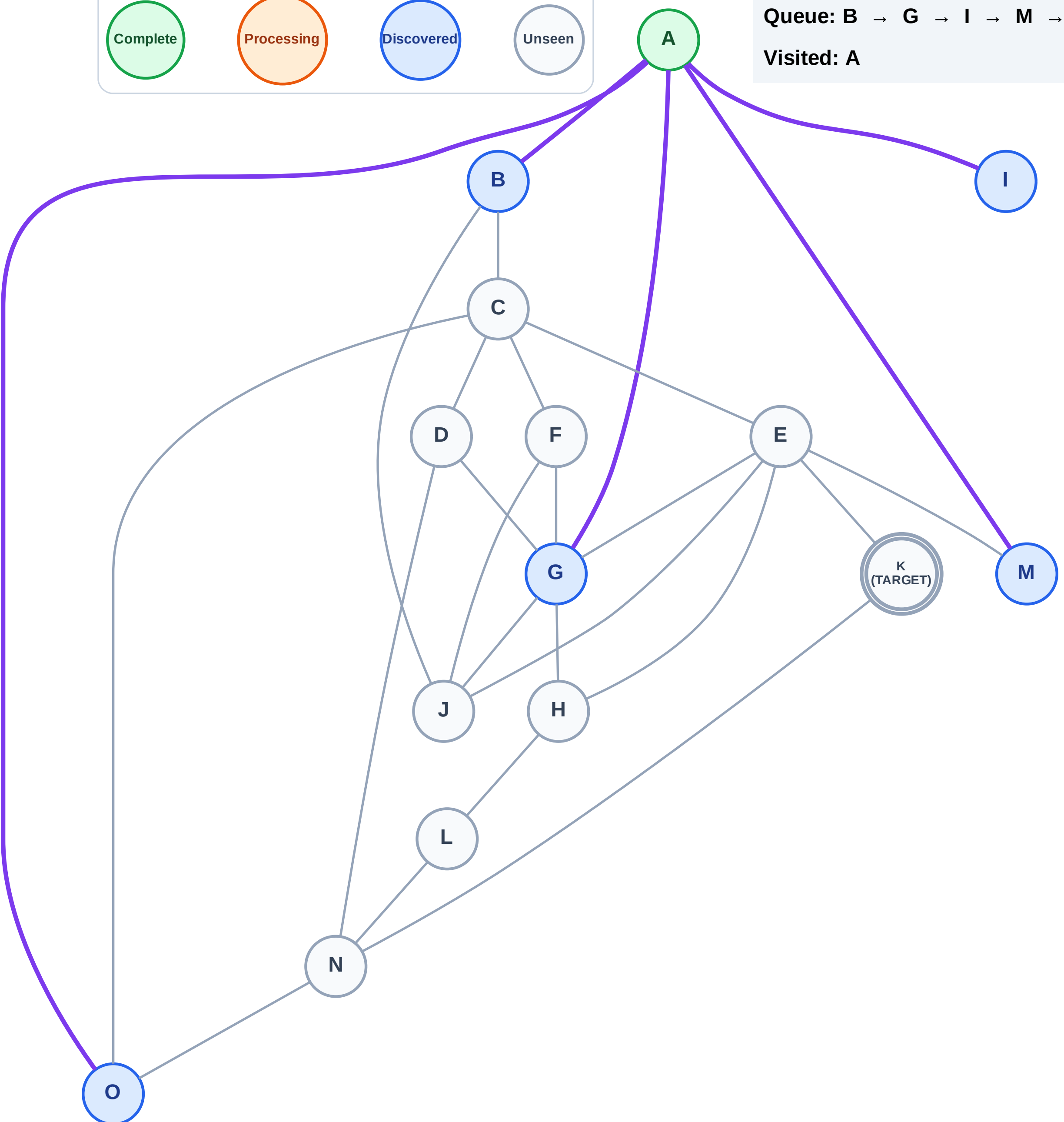
Step 3: Discover B, G, I, M, O from A

## Legend



Queue: B → G → I → M → O

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

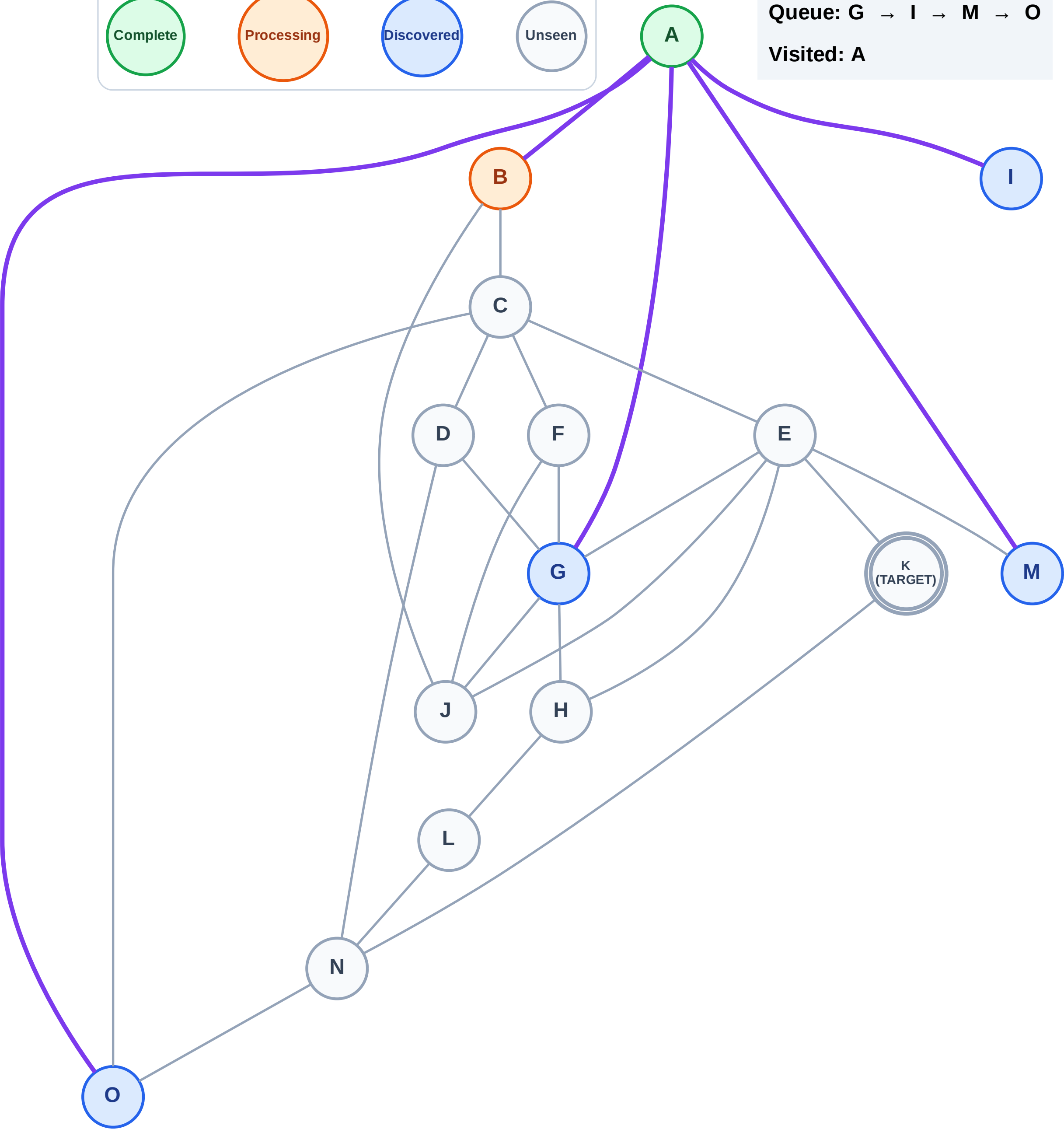
Processing

Discovered

Unseen

Queue: G → I → M → O

Visited: A



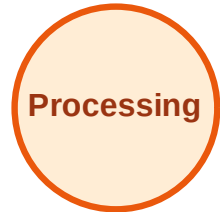
# BFS Traversal

Step 5: Discover C, J from B

## Legend



Complete



Processing



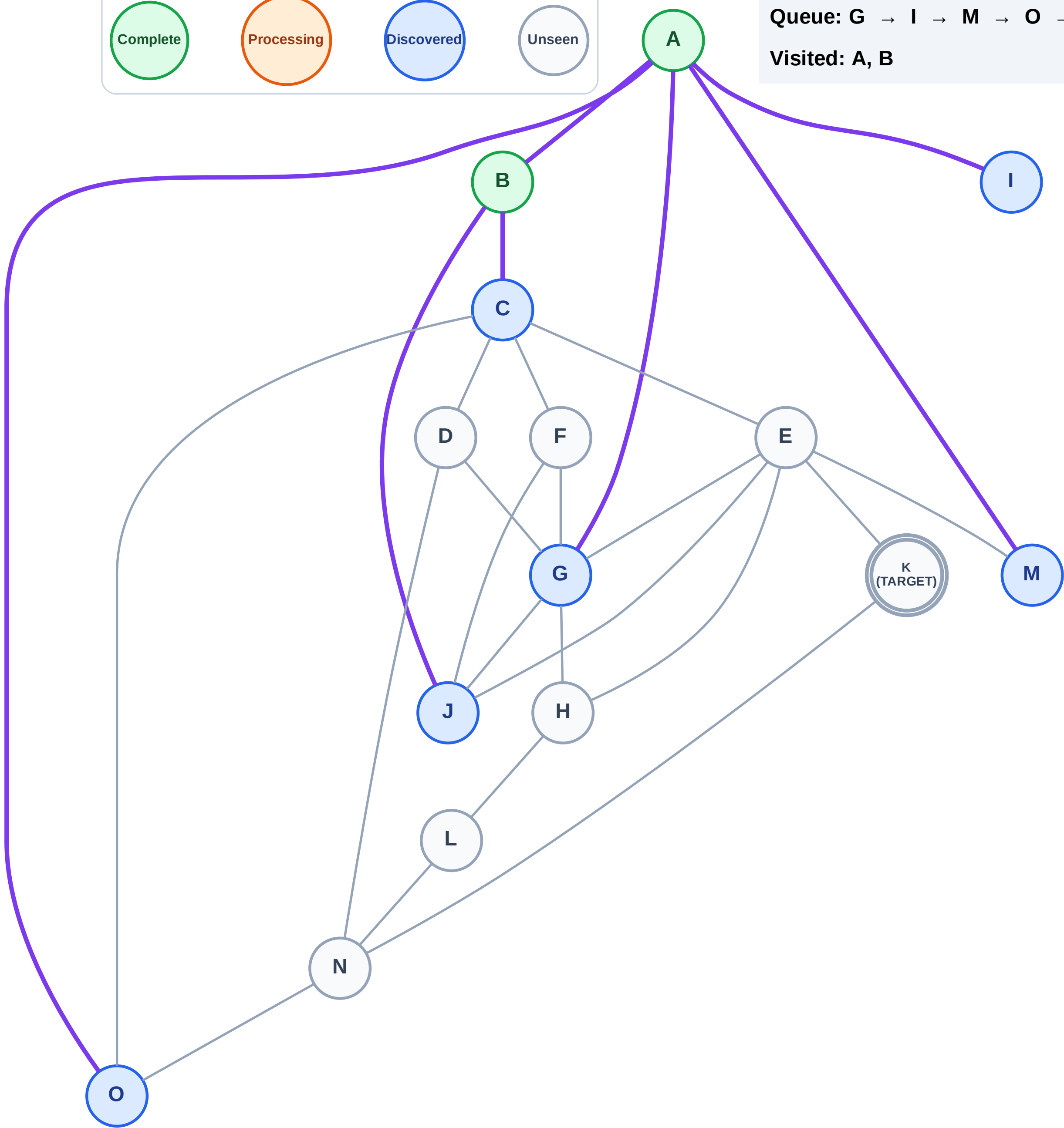
Discovered



Unseen

Queue: G → I → M → O → C → J

Visited: A, B





# BFS Traversal

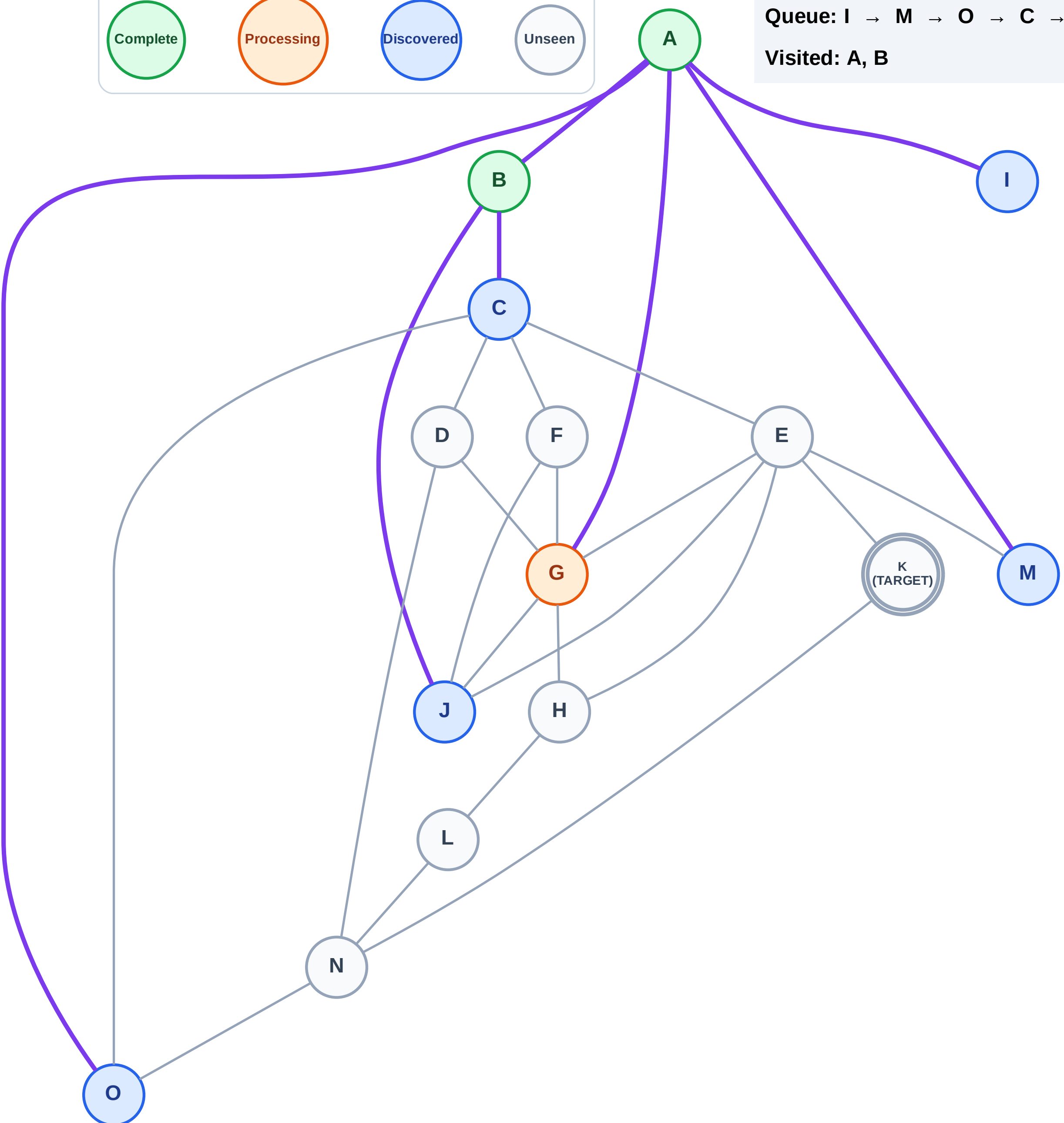
Step 6: Process node G

## Legend



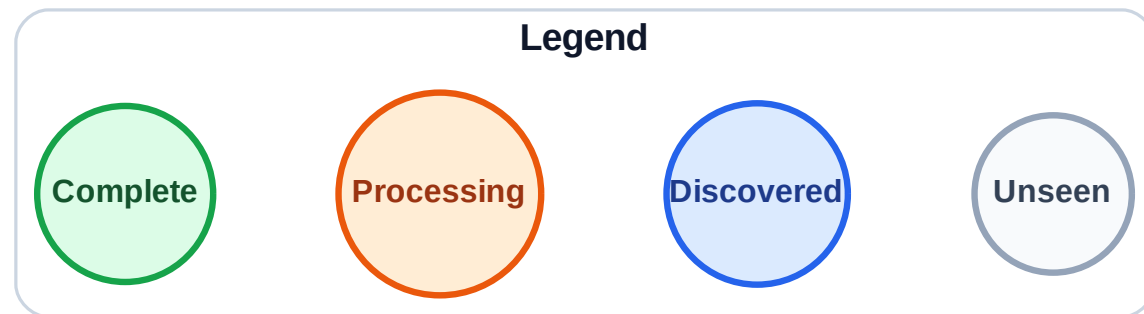
Queue: I → M → O → C → J

Visited: A, B

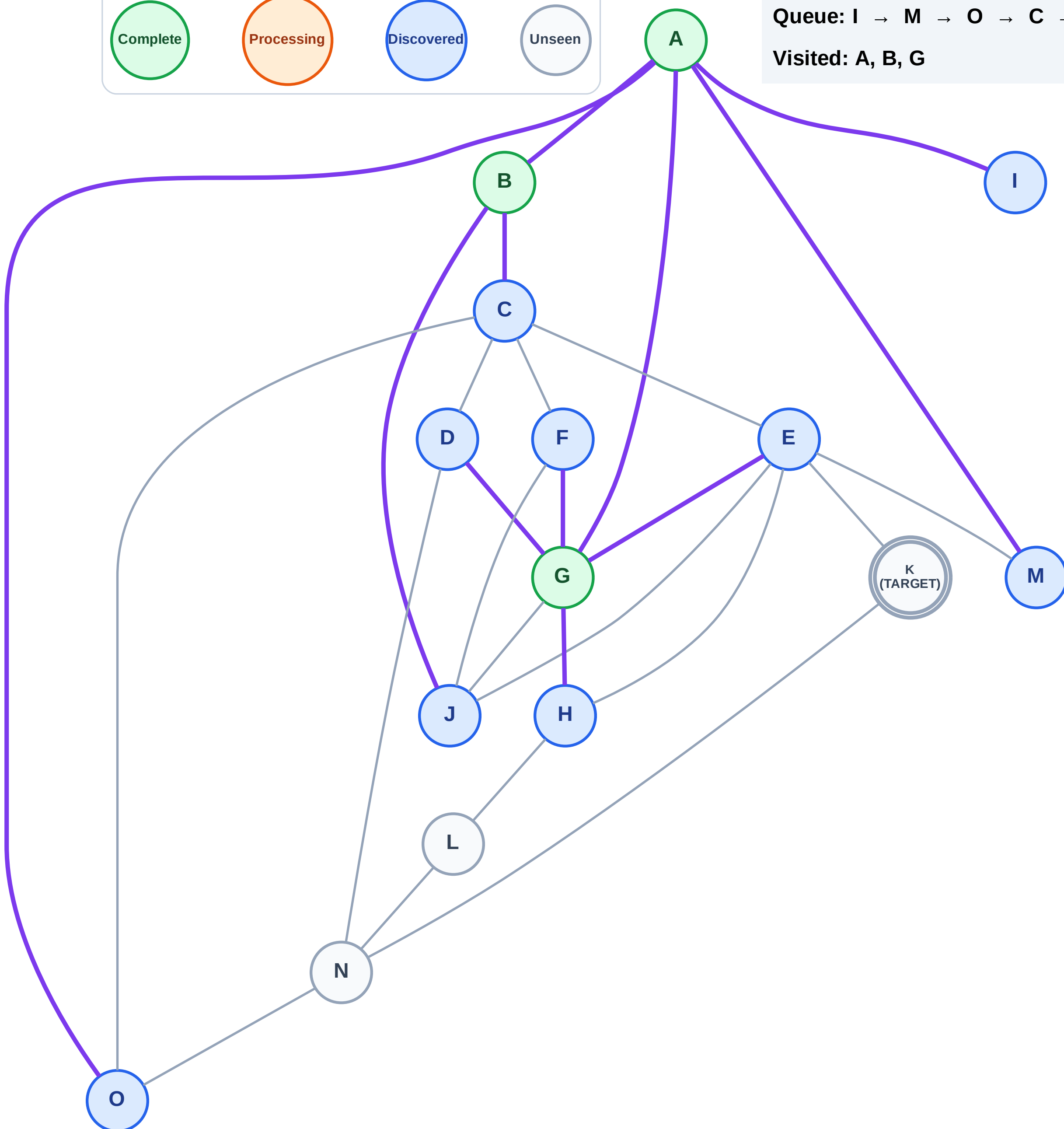


# BFS Traversal

Step 7: Discover D, E, F, H from G



Queue: I → M → O → C → J → D → E → F → H  
Visited: A, B, G



# BFS Traversal

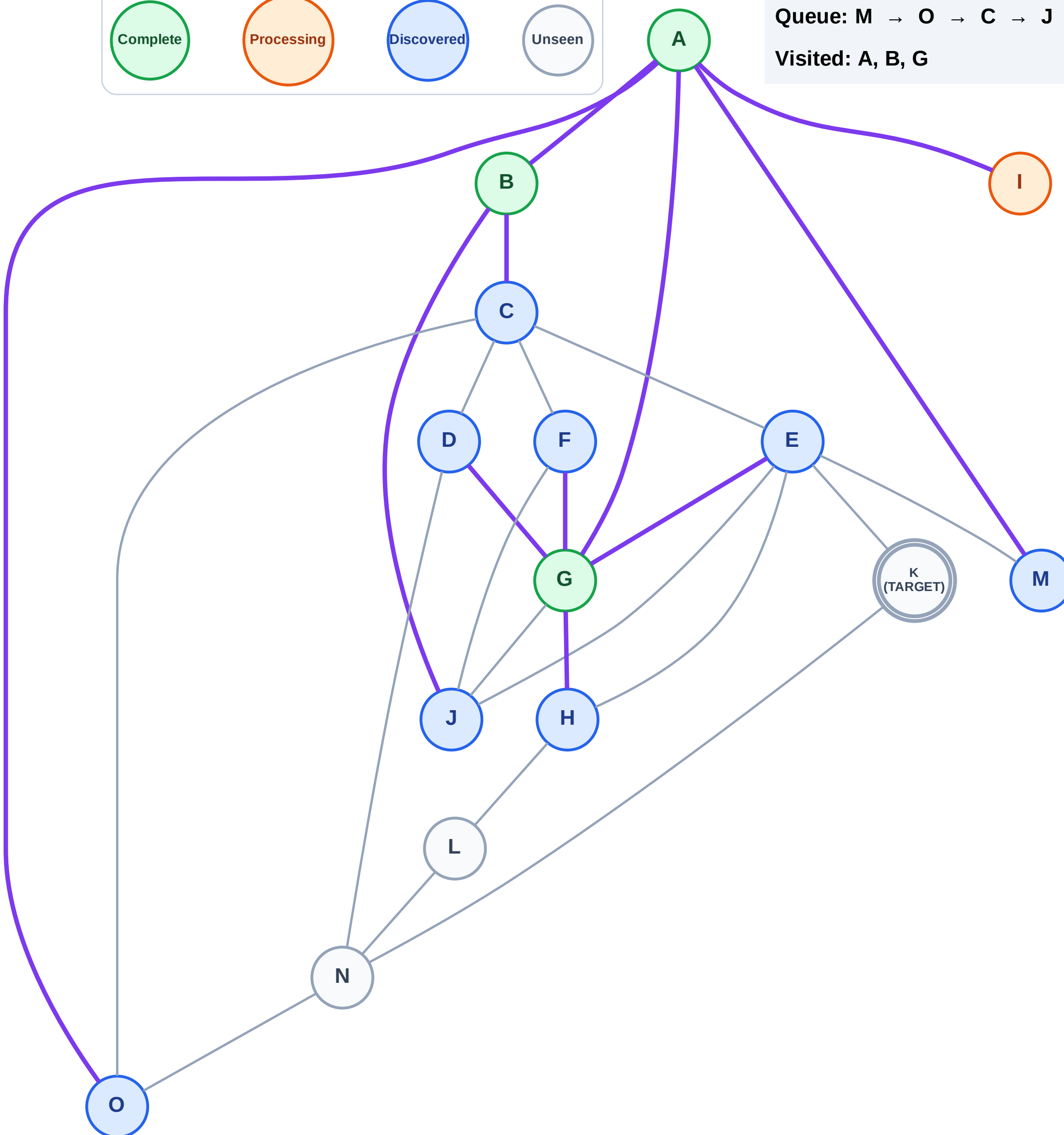
Step 8: Process node I

## Legend



Queue: M → O → C → J → D → E → F → H

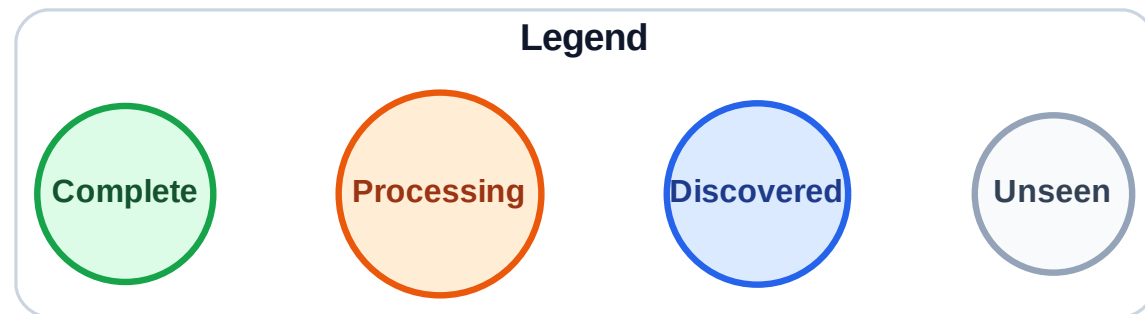
Visited: A, B, G



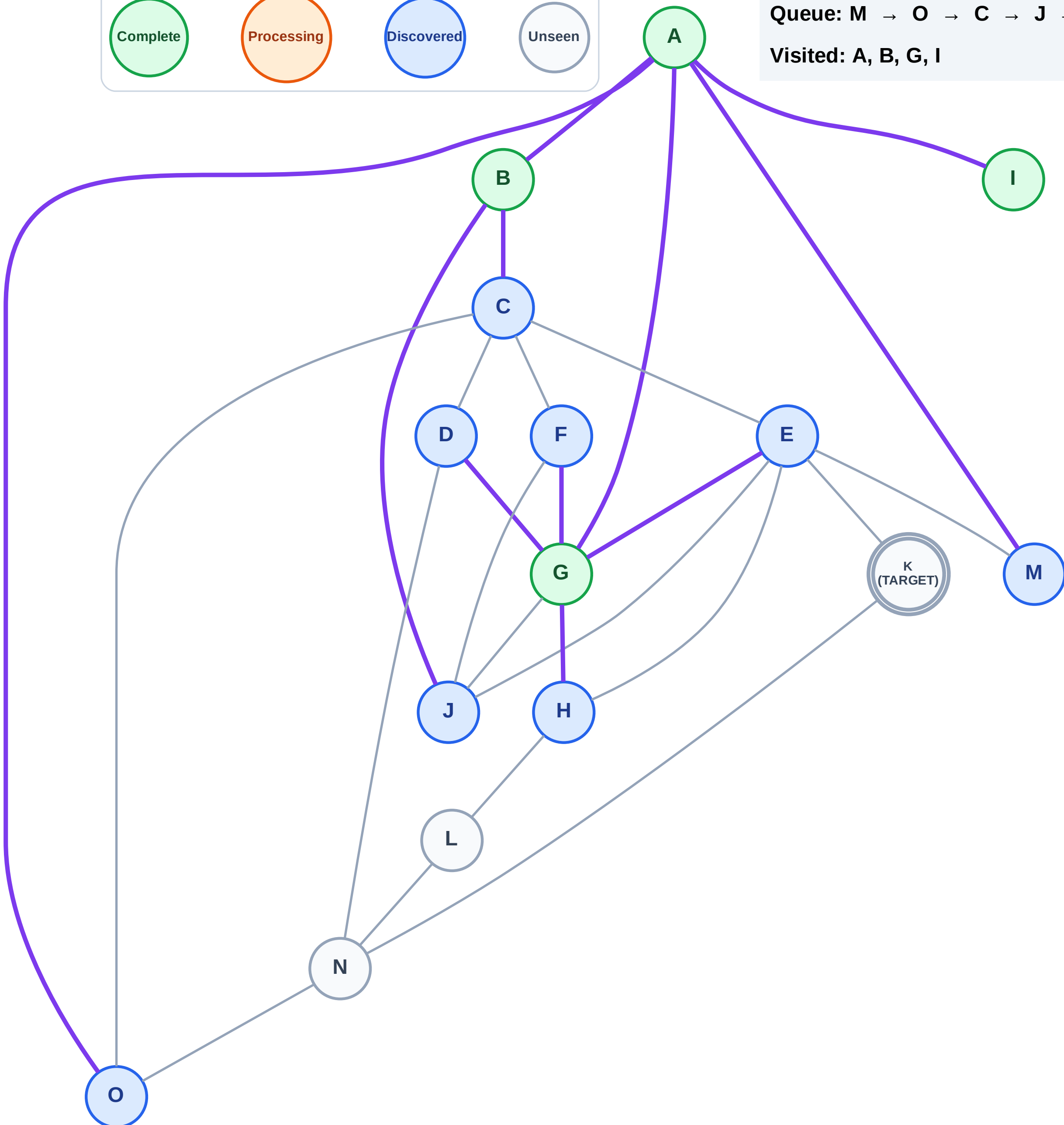


# BFS Traversal

Step 9: No new neighbors from I



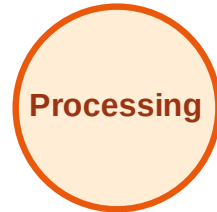
Queue: M → O → C → J → D → E → F → H  
Visited: A, B, G, I



# BFS Traversal

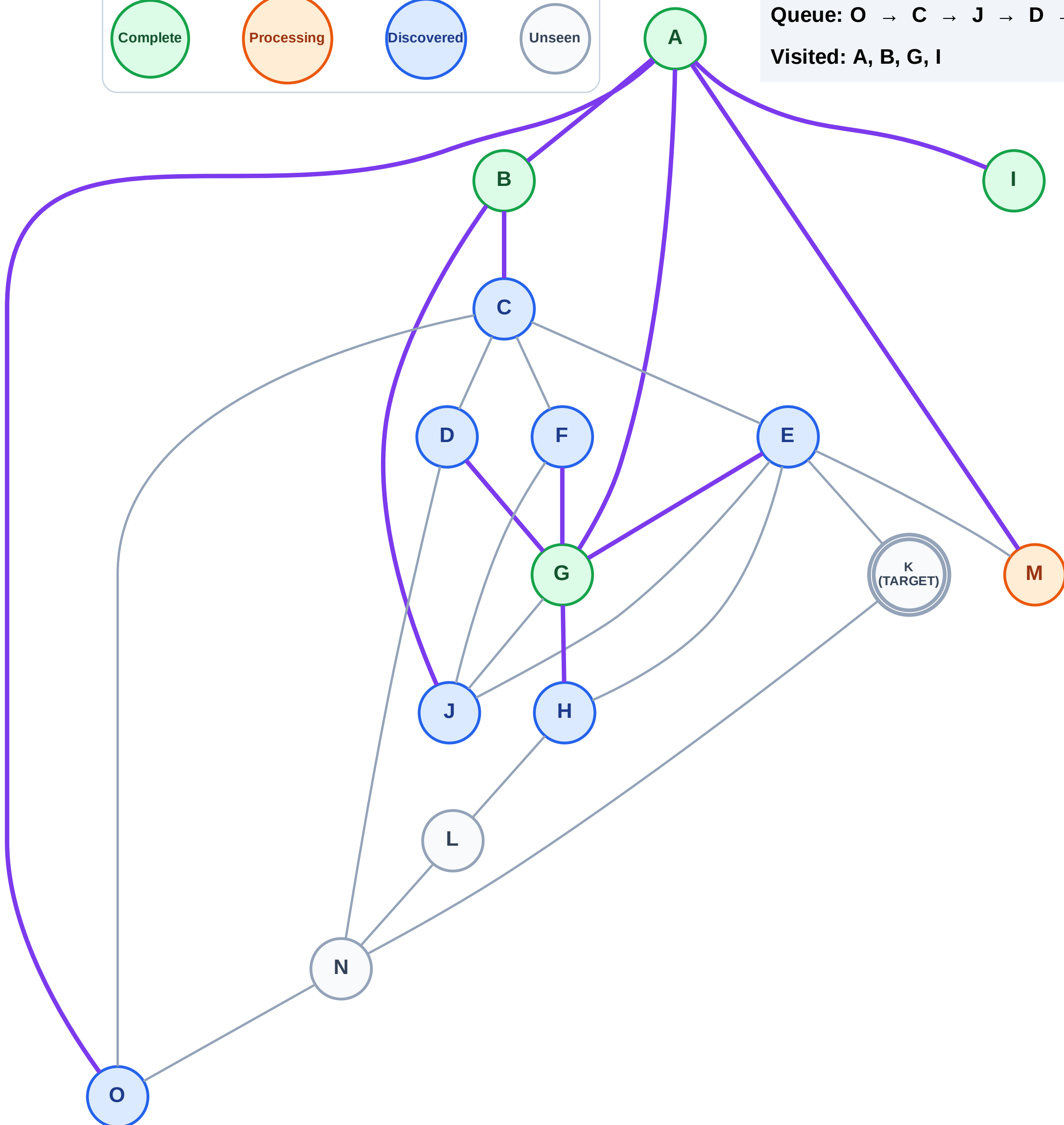
Step 10: Process node M

## Legend



Queue: O → C → J → D → E → F → H

Visited: A, B, G, I



# BFS Traversal

Step 11: No new neighbors from M

Legend

Complete

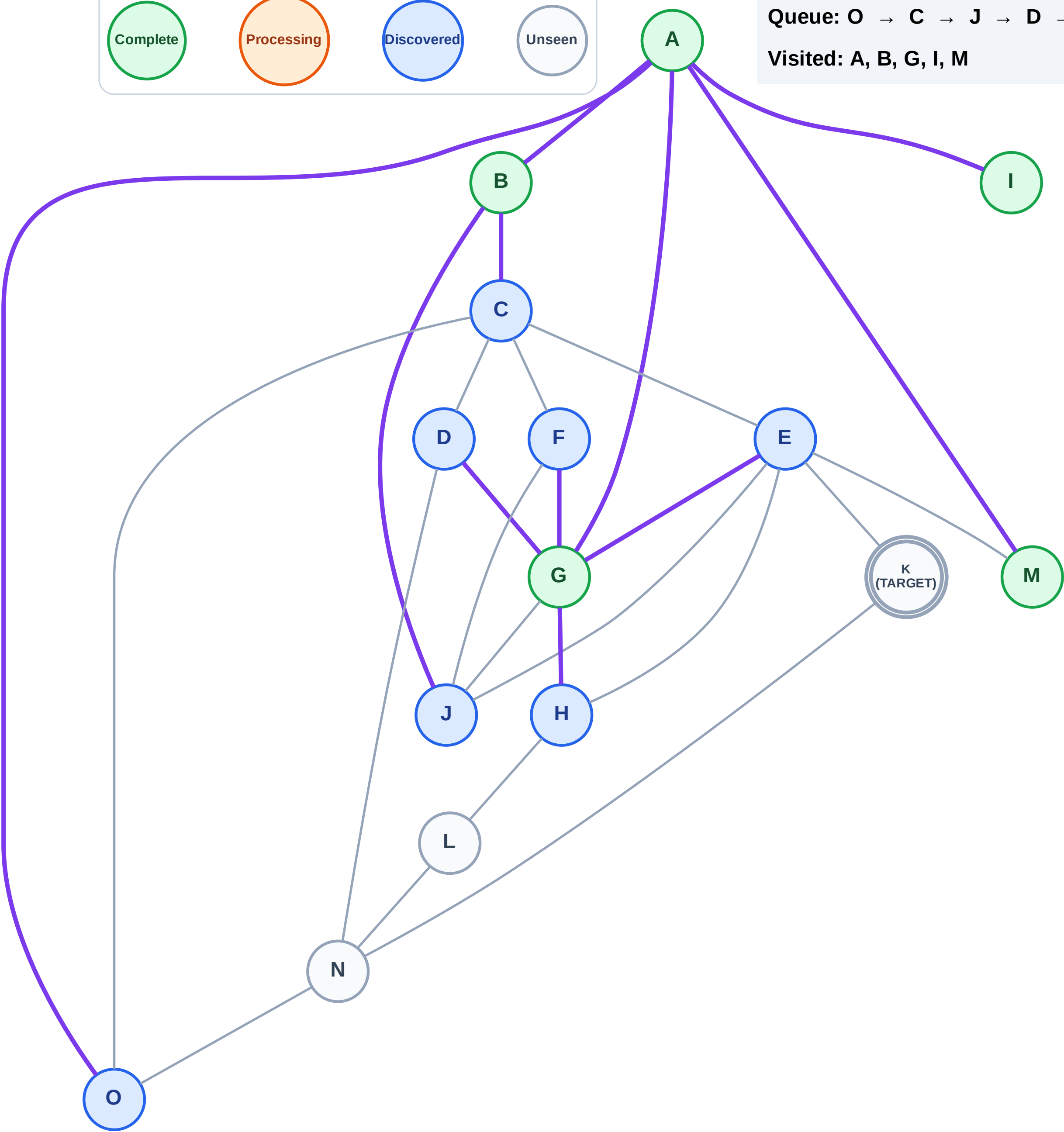
Processing

Discovered

Unseen

Queue: O → C → J → D → E → F → H

Visited: A, B, G, I, M



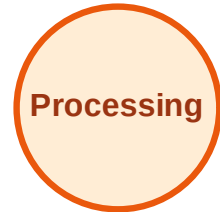
# BFS Traversal

Step 12: Process node O

## Legend



Complete



Processing



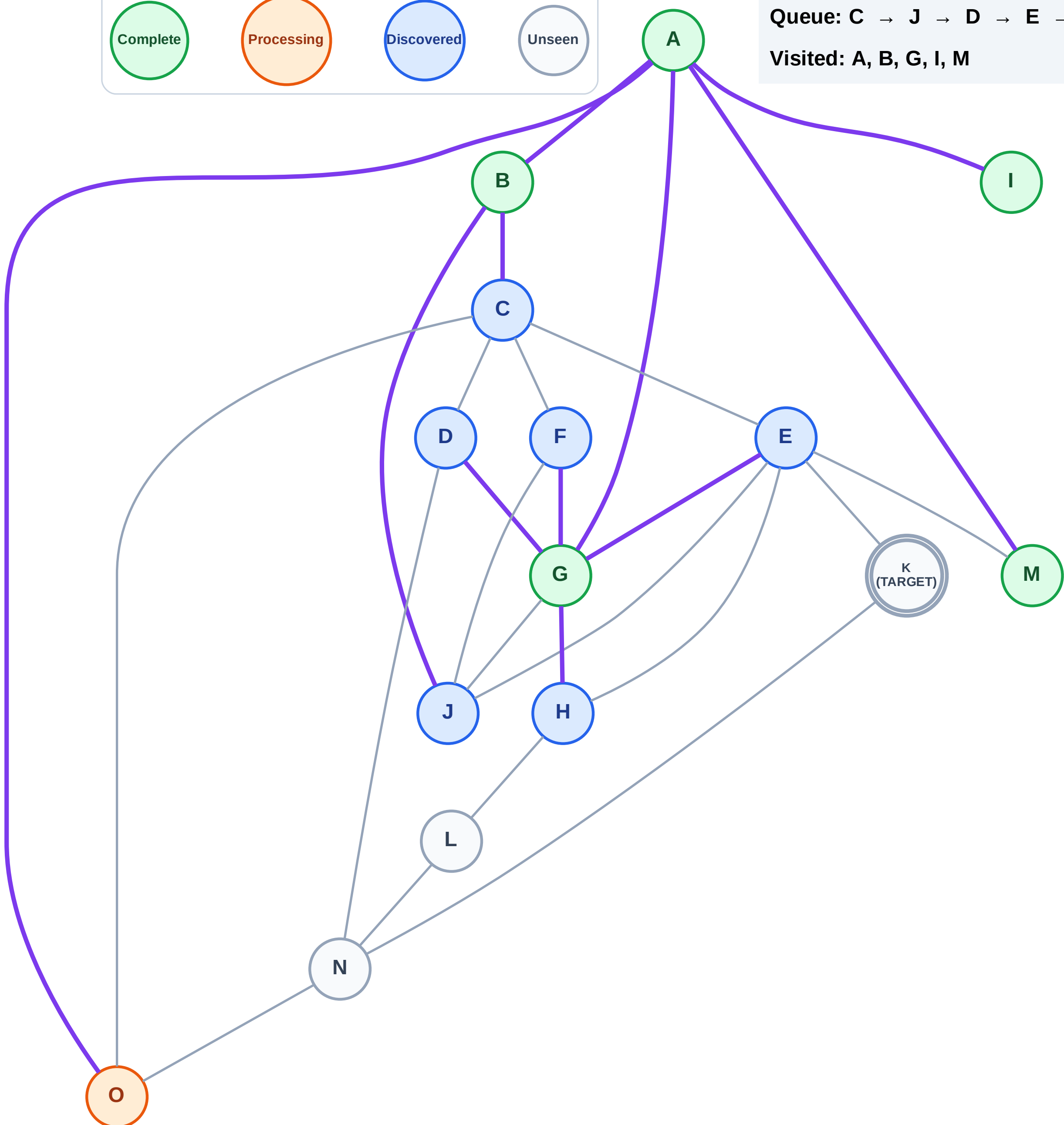
Discovered



Unseen

Queue: C → J → D → E → F → H

Visited: A, B, G, I, M

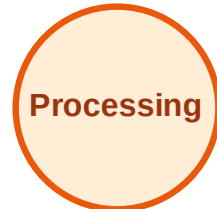




# BFS Traversal

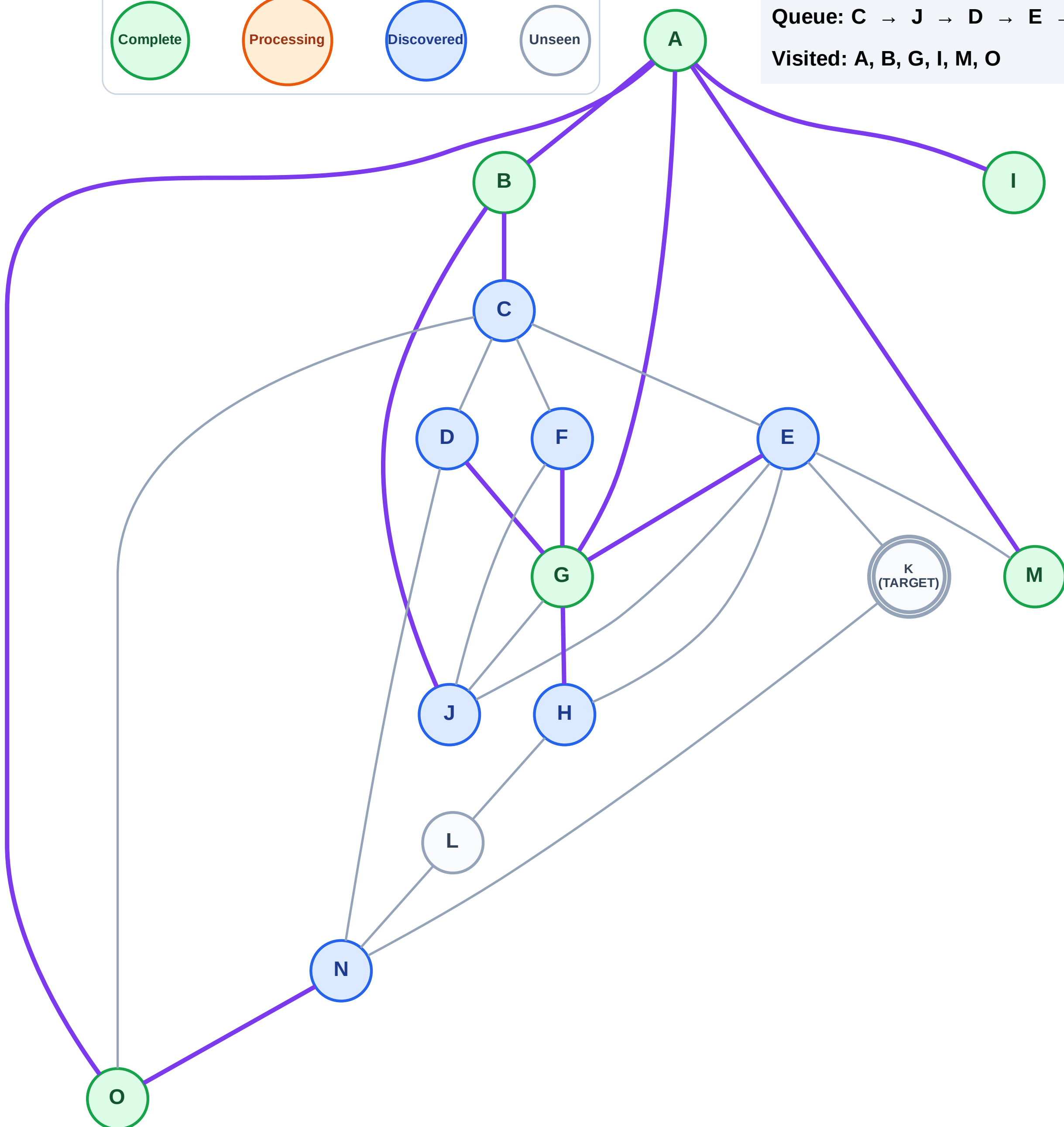
Step 13: Discover N from O

## Legend



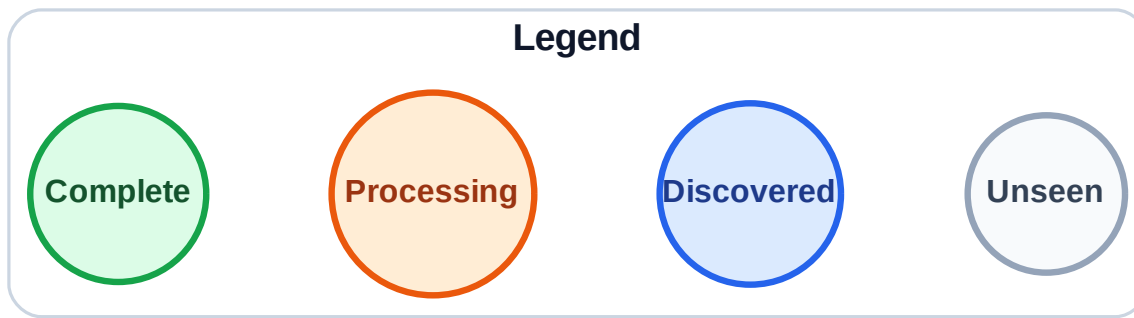
Queue: C → J → D → E → F → H → N

Visited: A, B, G, I, M, O



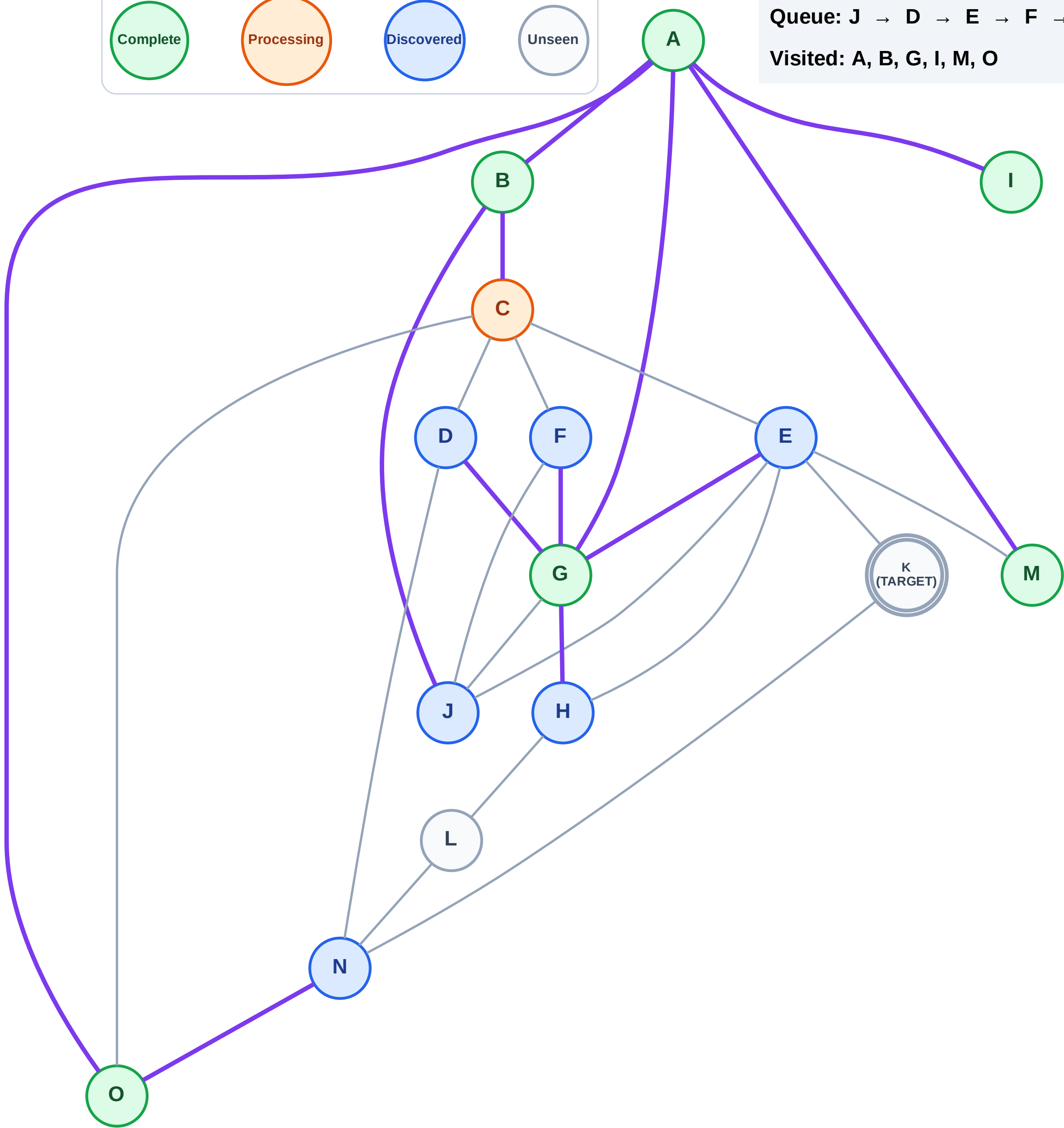
# BFS Traversal

Step 14: Process node C



Queue: J → D → E → F → H → N

Visited: A, B, G, I, M, O



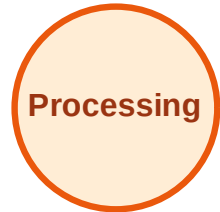
# BFS Traversal

Step 15: No new neighbors from C

## Legend



Complete



Processing



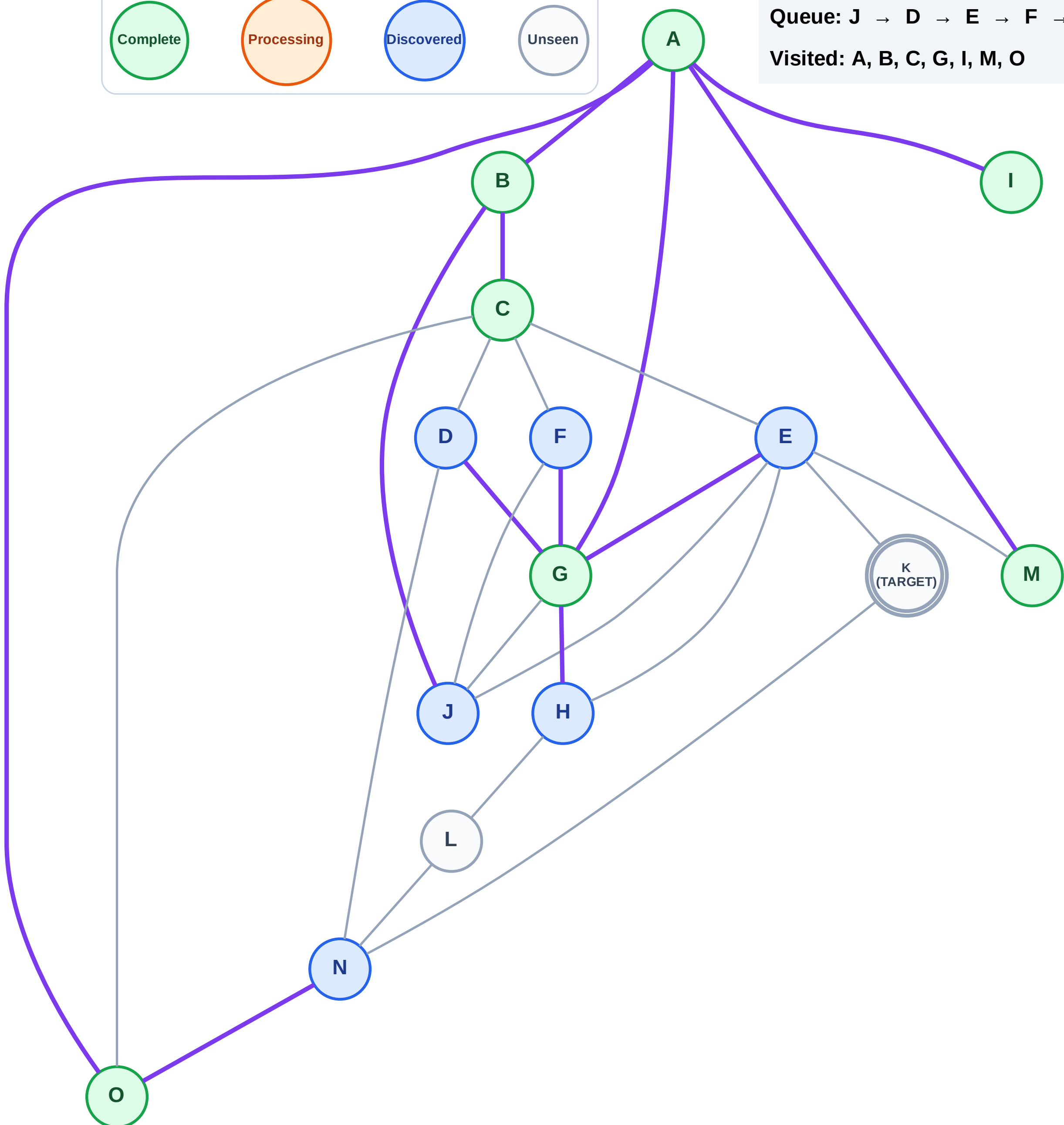
Discovered



Unseen

Queue: J → D → E → F → H → N

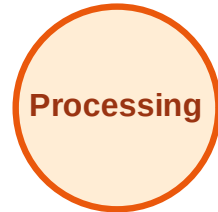
Visited: A, B, C, G, I, M, O



# BFS Traversal

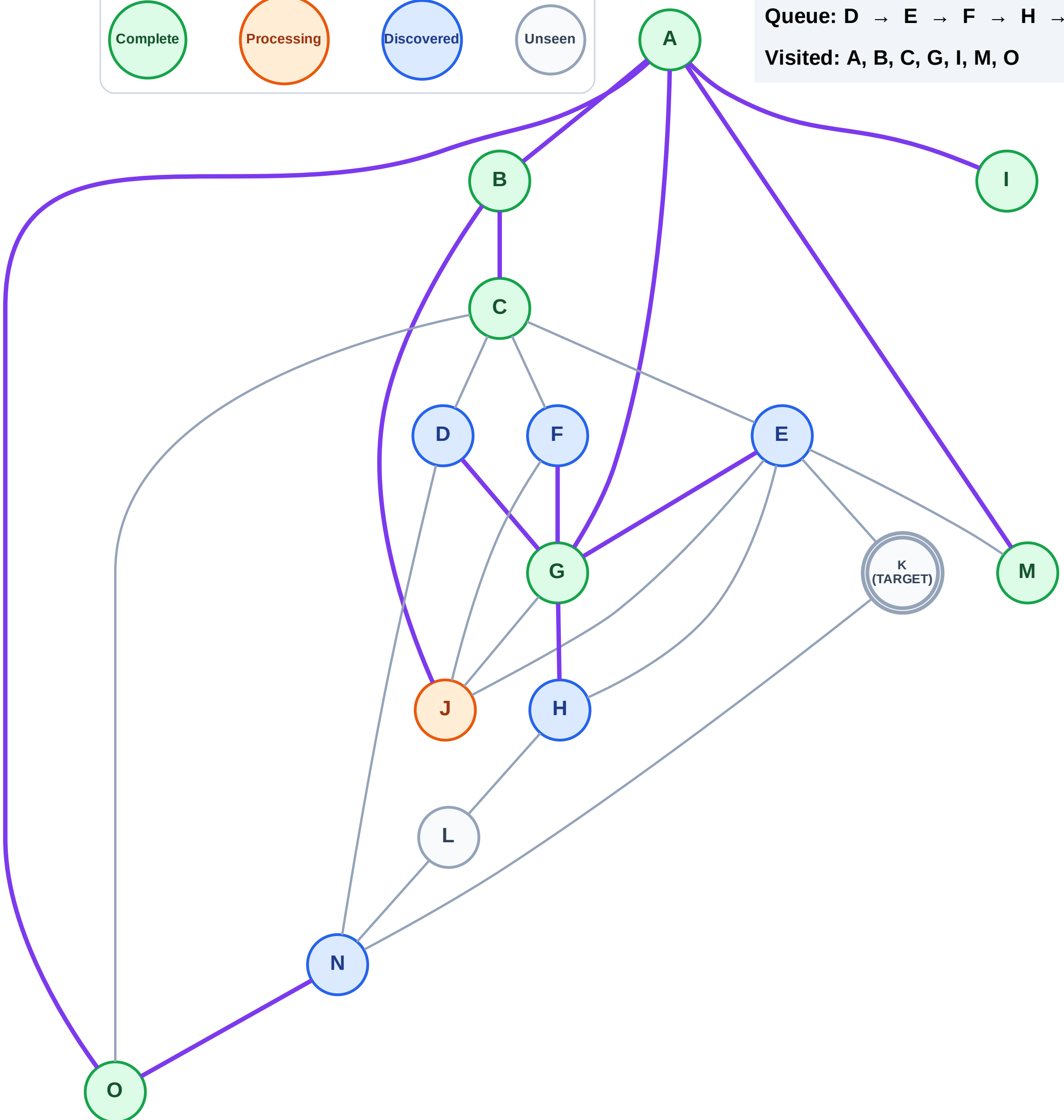
Step 16: Process node J

## Legend



Queue: D → E → F → H → N

Visited: A, B, C, G, I, M, O

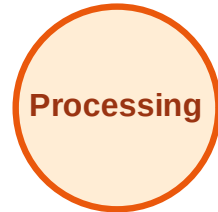




# BFS Traversal

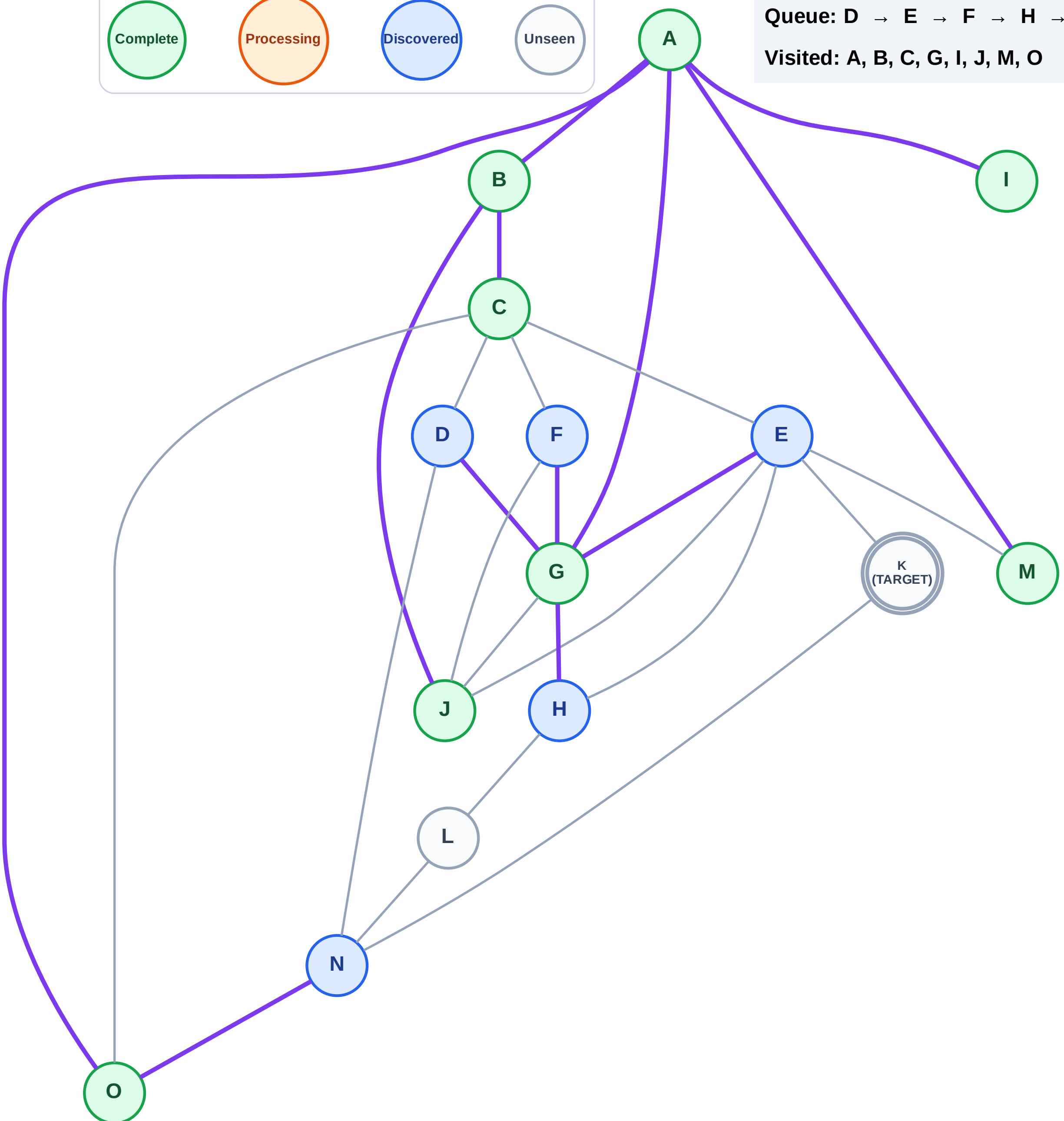
Step 17: No new neighbors from J

## Legend



Queue: D → E → F → H → N

Visited: A, B, C, G, I, J, M, O



# BFS Traversal

Step 18: Process node D

Legend

Complete

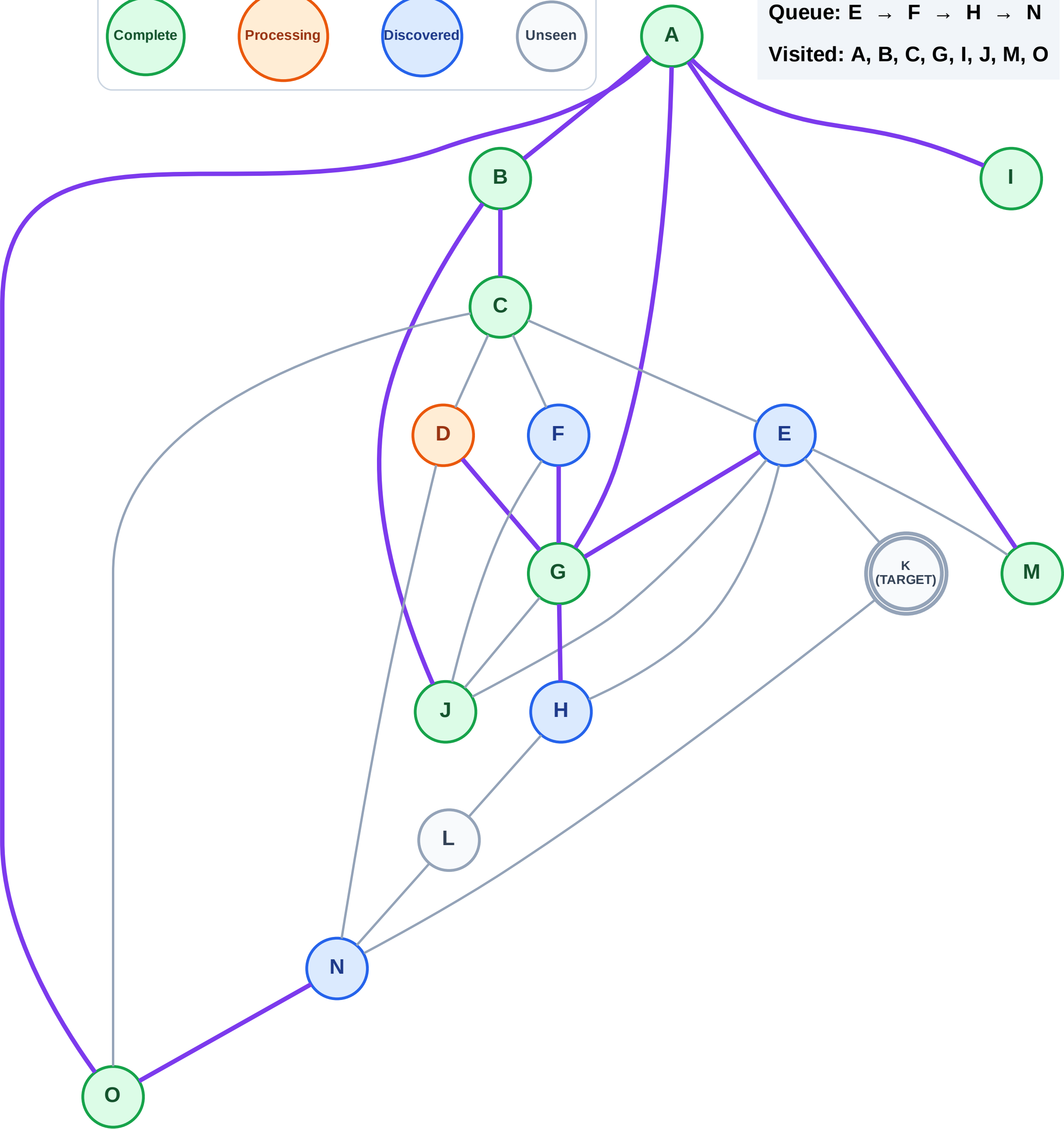
Processing

Discovered

Unseen

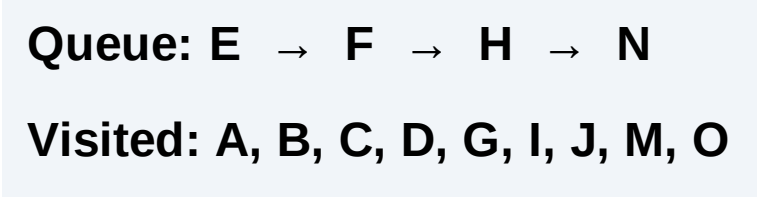
Queue: E → F → H → N

Visited: A, B, C, G, I, J, M, O



Step 19: No new neighbors from D

Step 19: No new neighbors from D



# BFS Traversal

Step 20: Process node E

Legend

Complete

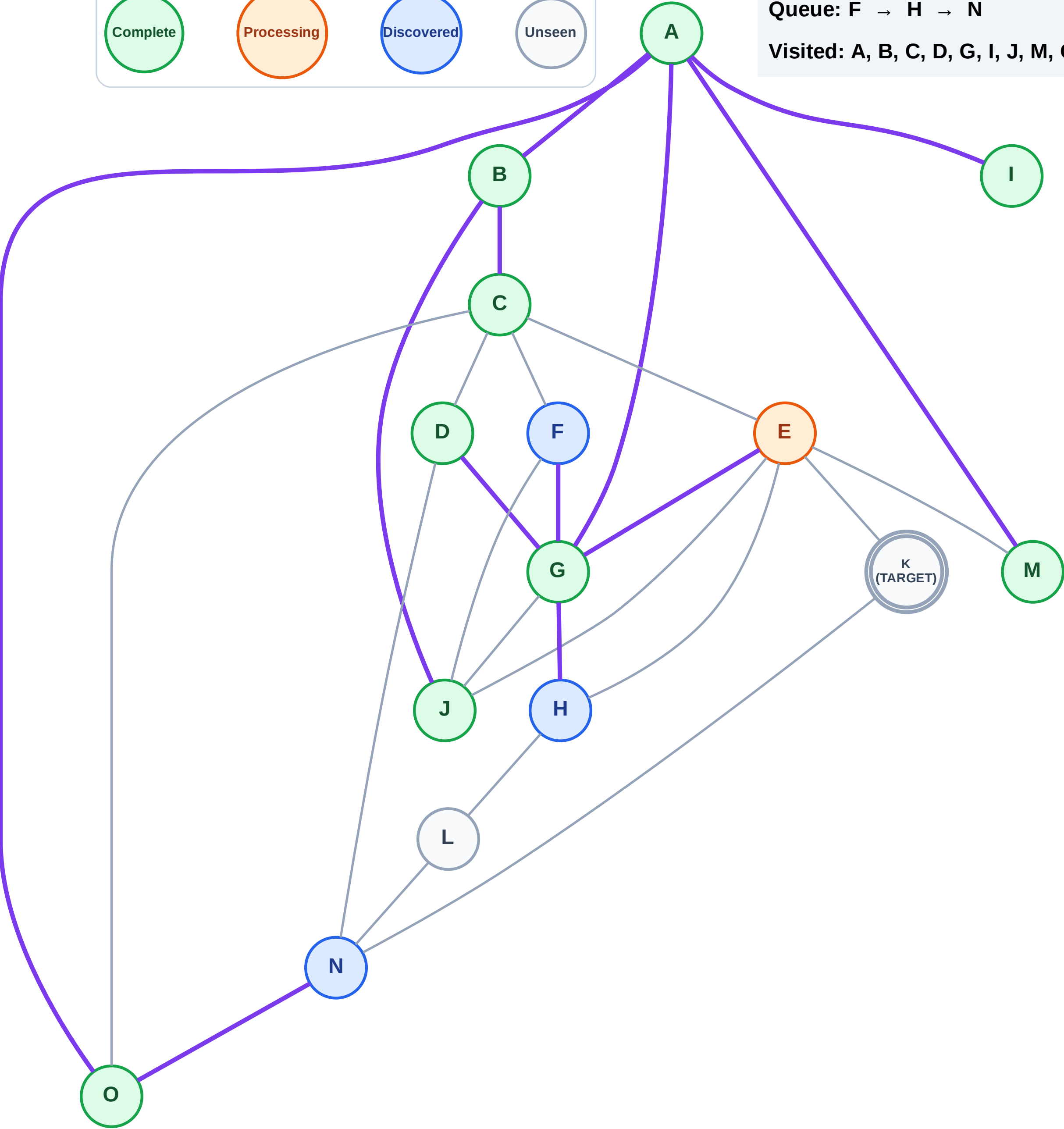
Processing

Discovered

Unseen

Queue: F → H → N

Visited: A, B, C, D, G, I, J, M, O



# BFS Traversal

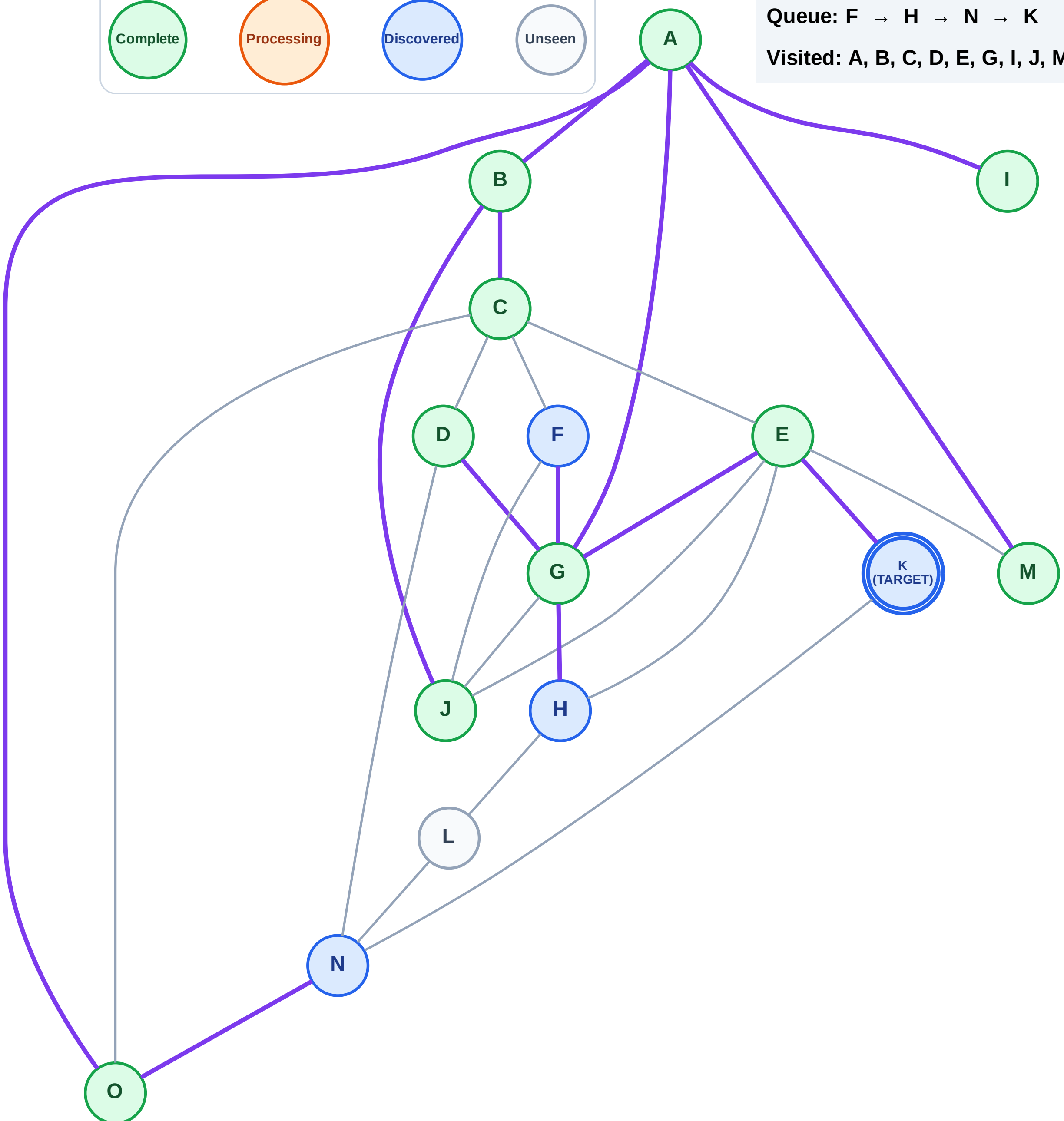
Step 21: Discover K from E

## Legend



Queue: F → H → N → K

Visited: A, B, C, D, E, G, I, J, M, O

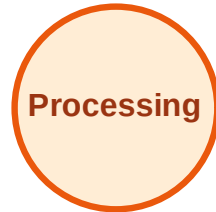




# BFS Traversal

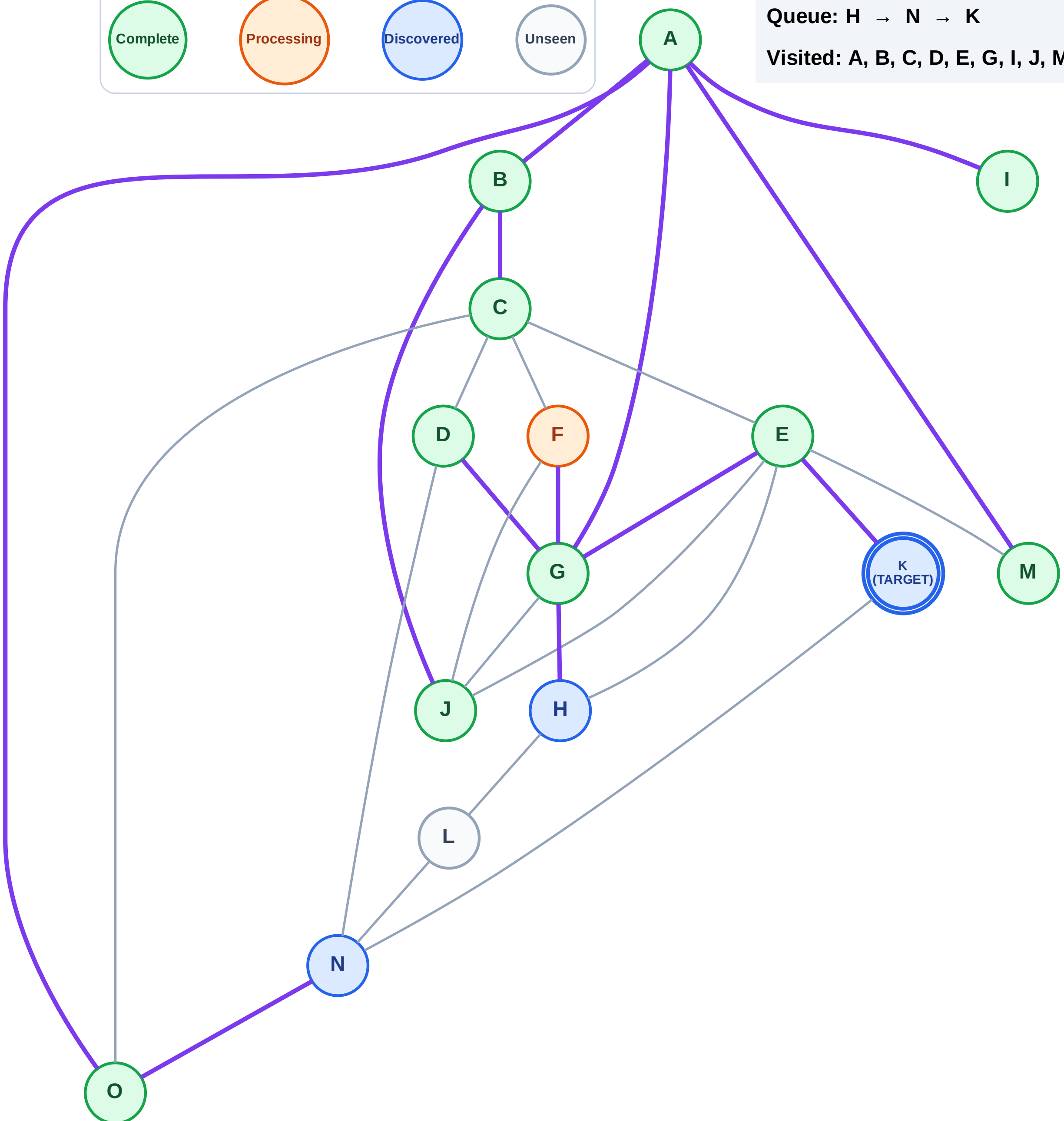
Step 22: Process node F

## Legend



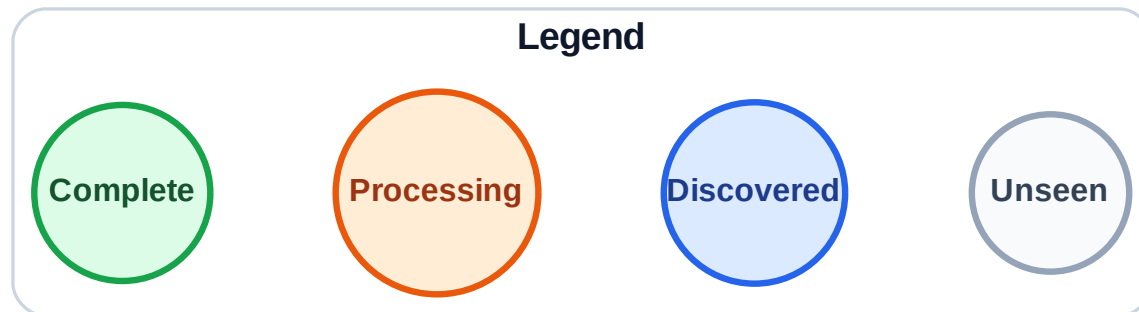
Queue: H → N → K

Visited: A, B, C, D, E, G, I, J, M, O



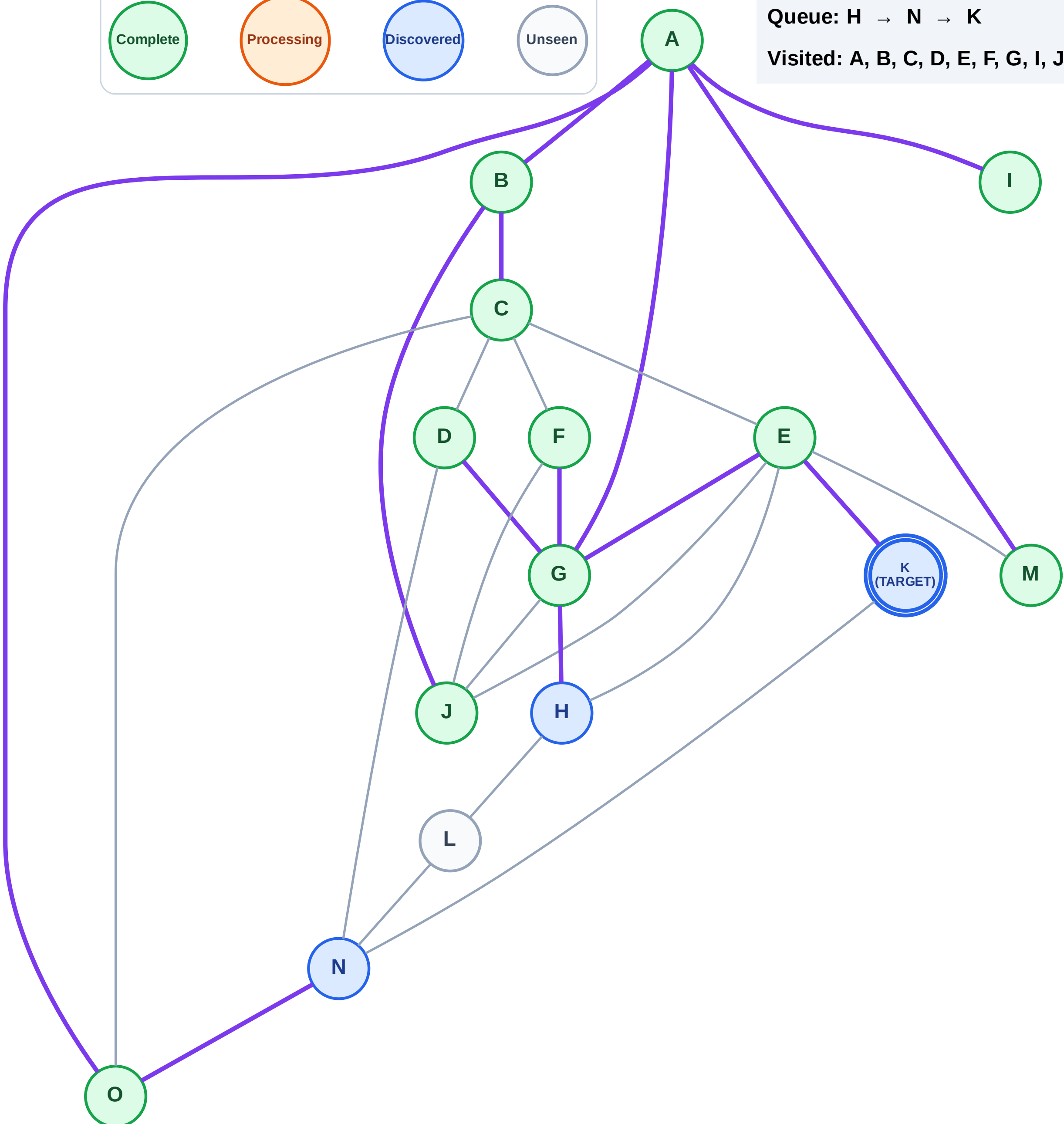
# BFS Traversal

Step 23: No new neighbors from F



Queue: H → N → K

Visited: A, B, C, D, E, F, G, I, J, M, O



# BFS Traversal

Step 24: Process node H

## Legend

Complete

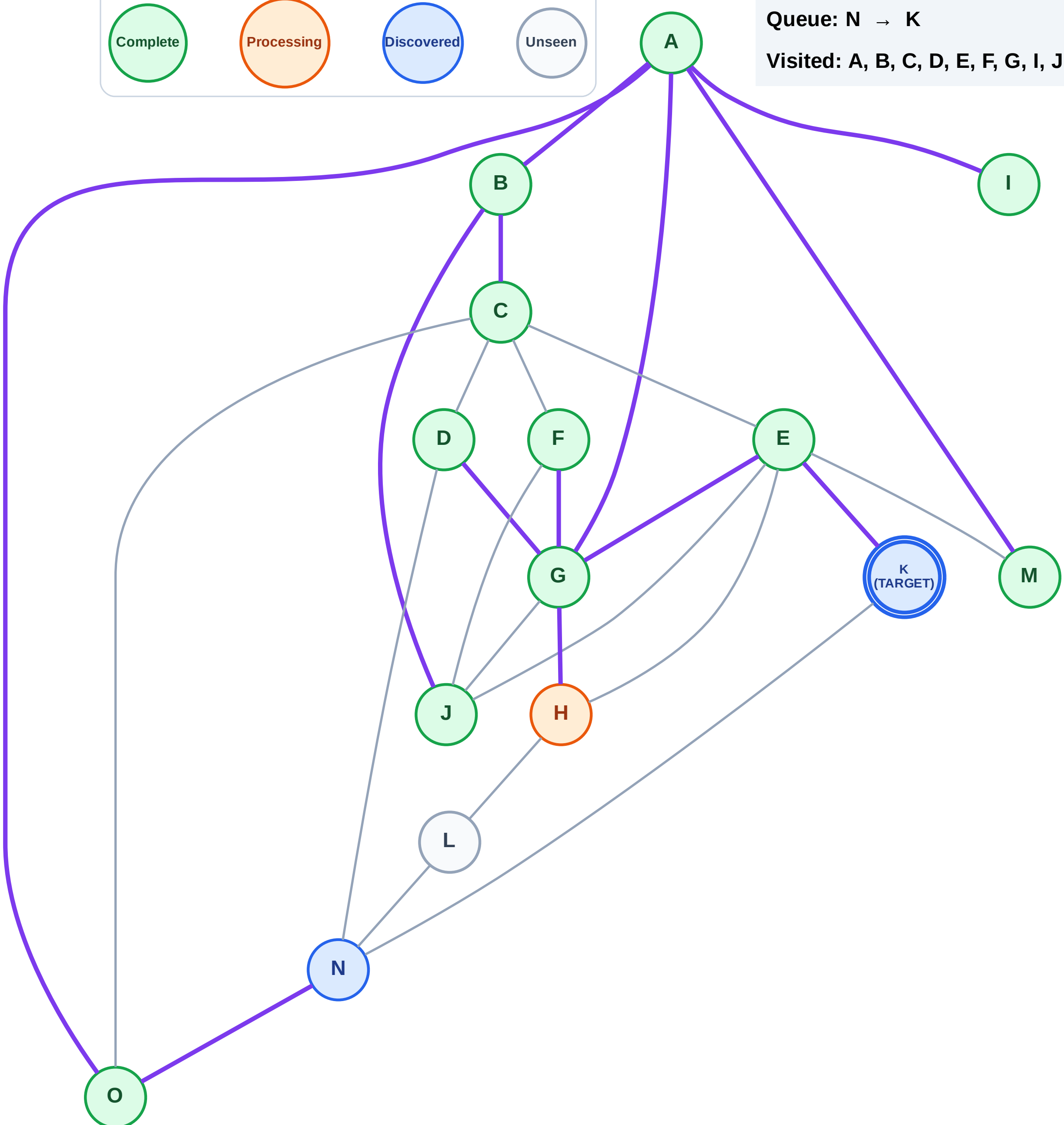
Processing

Discovered

Unseen

Queue: N → K

Visited: A, B, C, D, E, F, G, I, J, M, O





# BFS Traversal

Step 25: Discover L from H

Legend

Complete

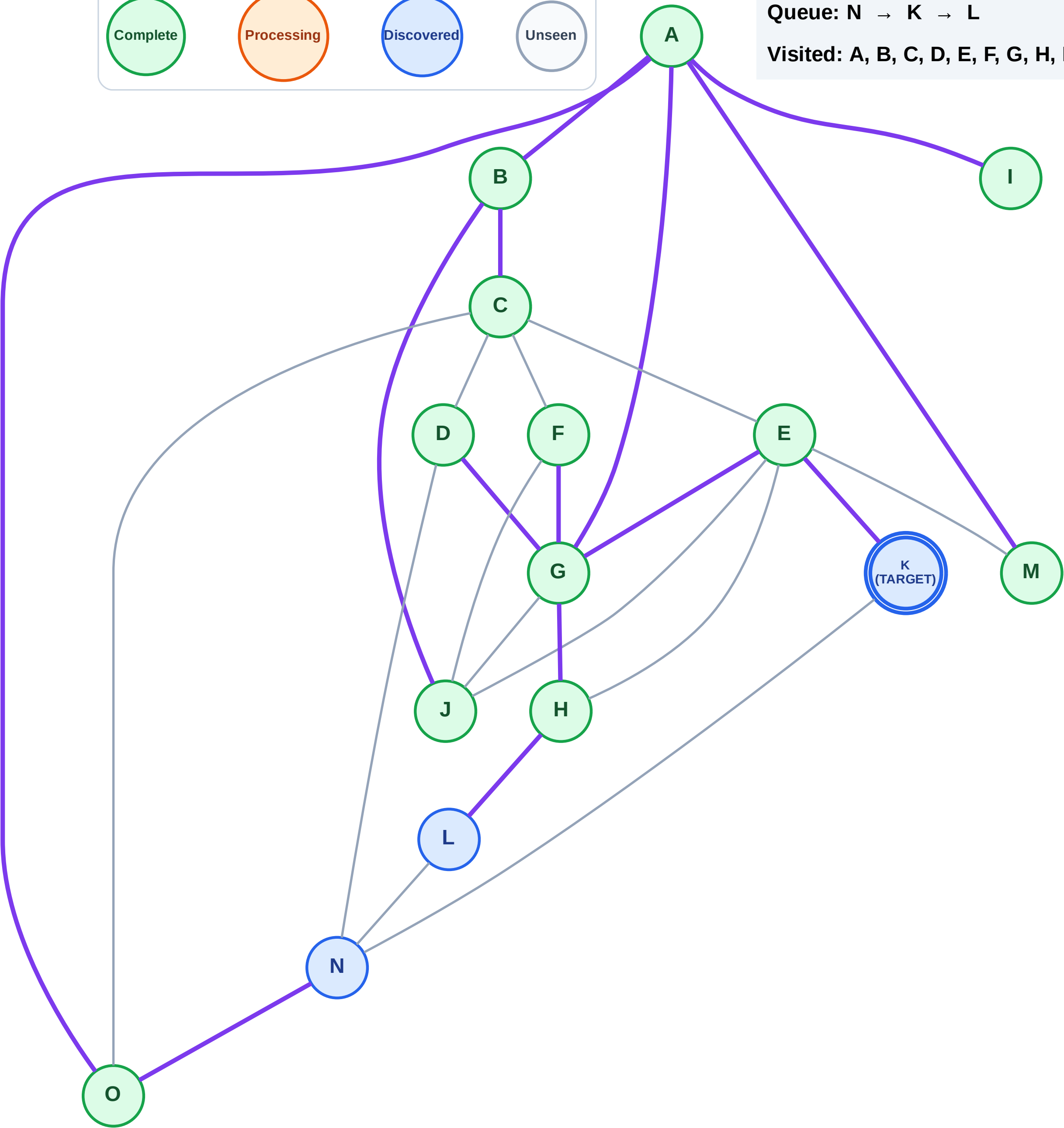
Processing

Discovered

Unseen

Queue: N → K → L

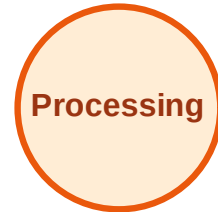
Visited: A, B, C, D, E, F, G, H, I, J, M, O



# BFS Traversal

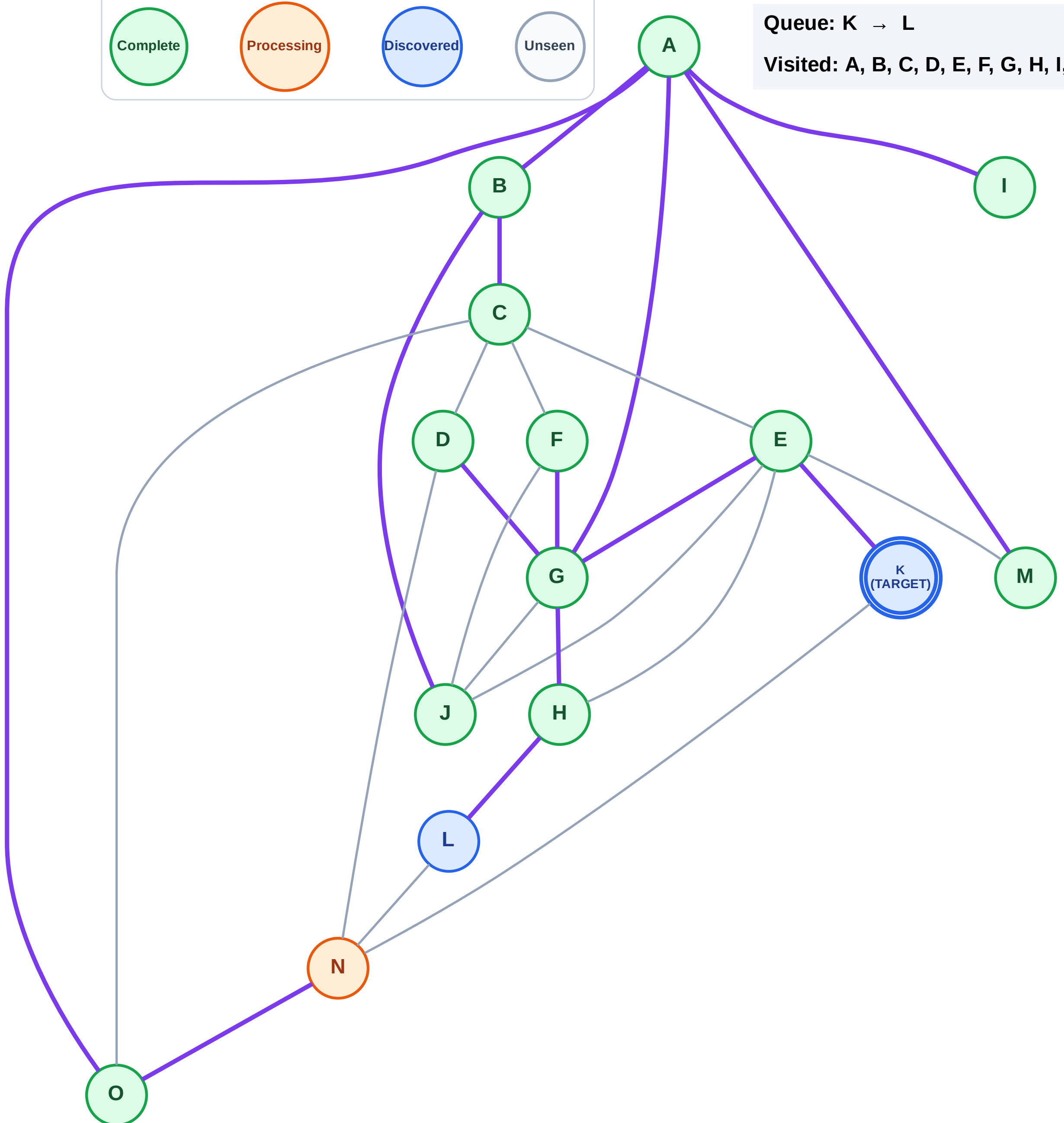
Step 26: Process node N

## Legend



Queue: K → L

Visited: A, B, C, D, E, F, G, H, I, J, M, O



# BFS Traversal

Step 27: No new neighbors from N

Legend

Complete

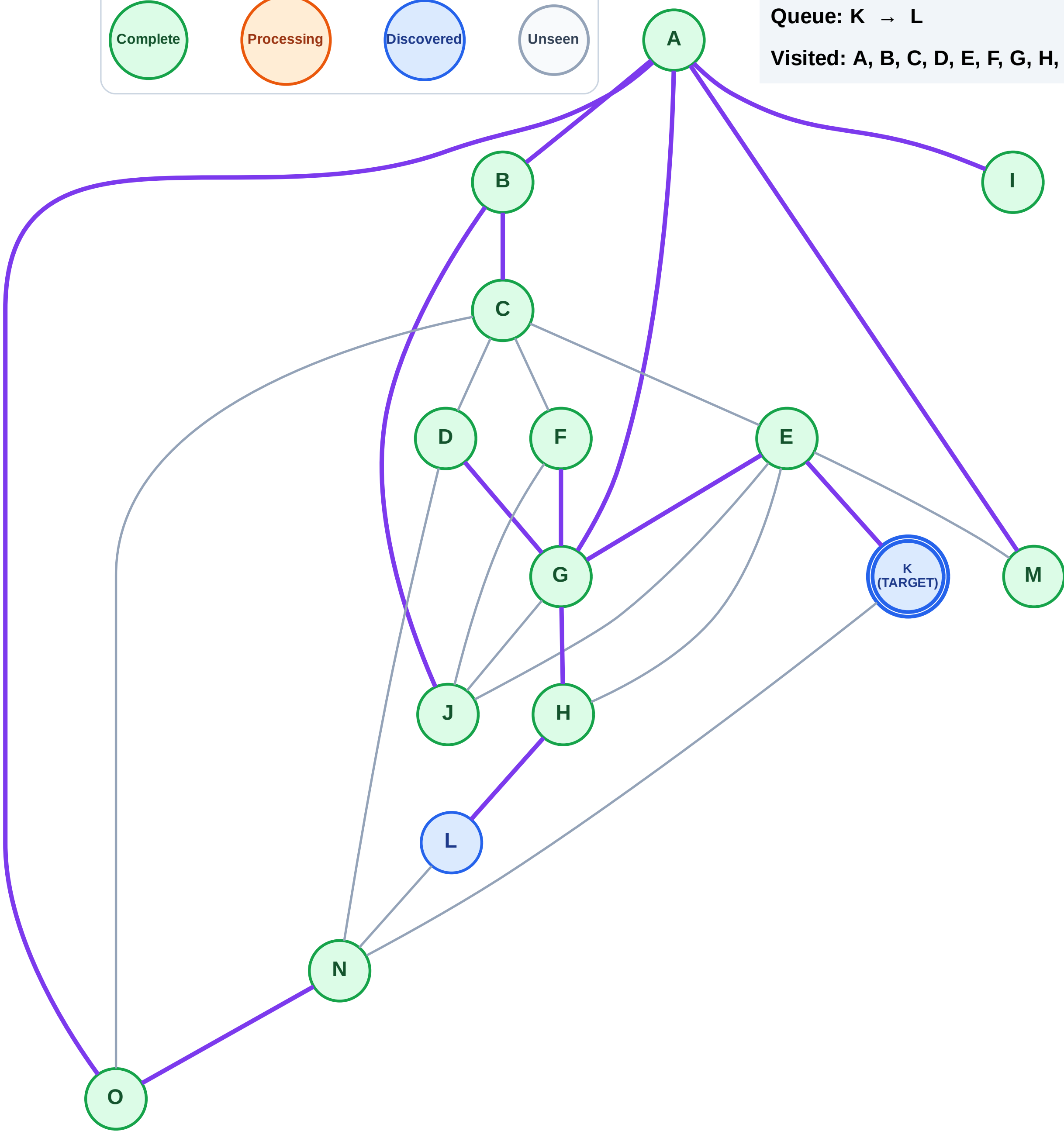
Processing

Discovered

Unseen

Queue: K → L

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O



# BFS Traversal

Step 28: Process node K

Legend

Complete

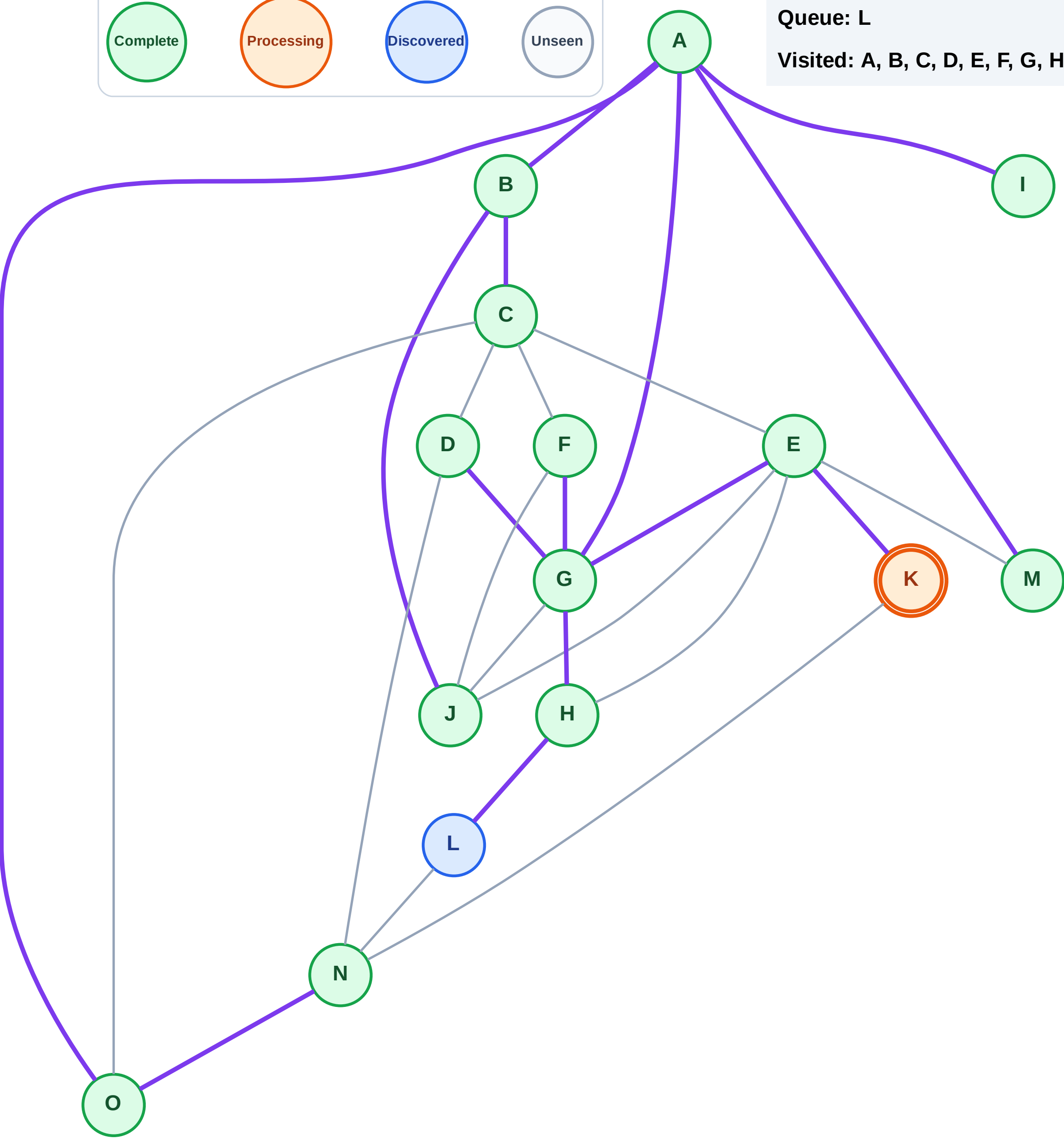
Processing

Discovered

Unseen

Queue: L

Visited: A, B, C, D, E, F, G, H, I, J, M, N, O





# BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → G → E → K

Legend

Complete

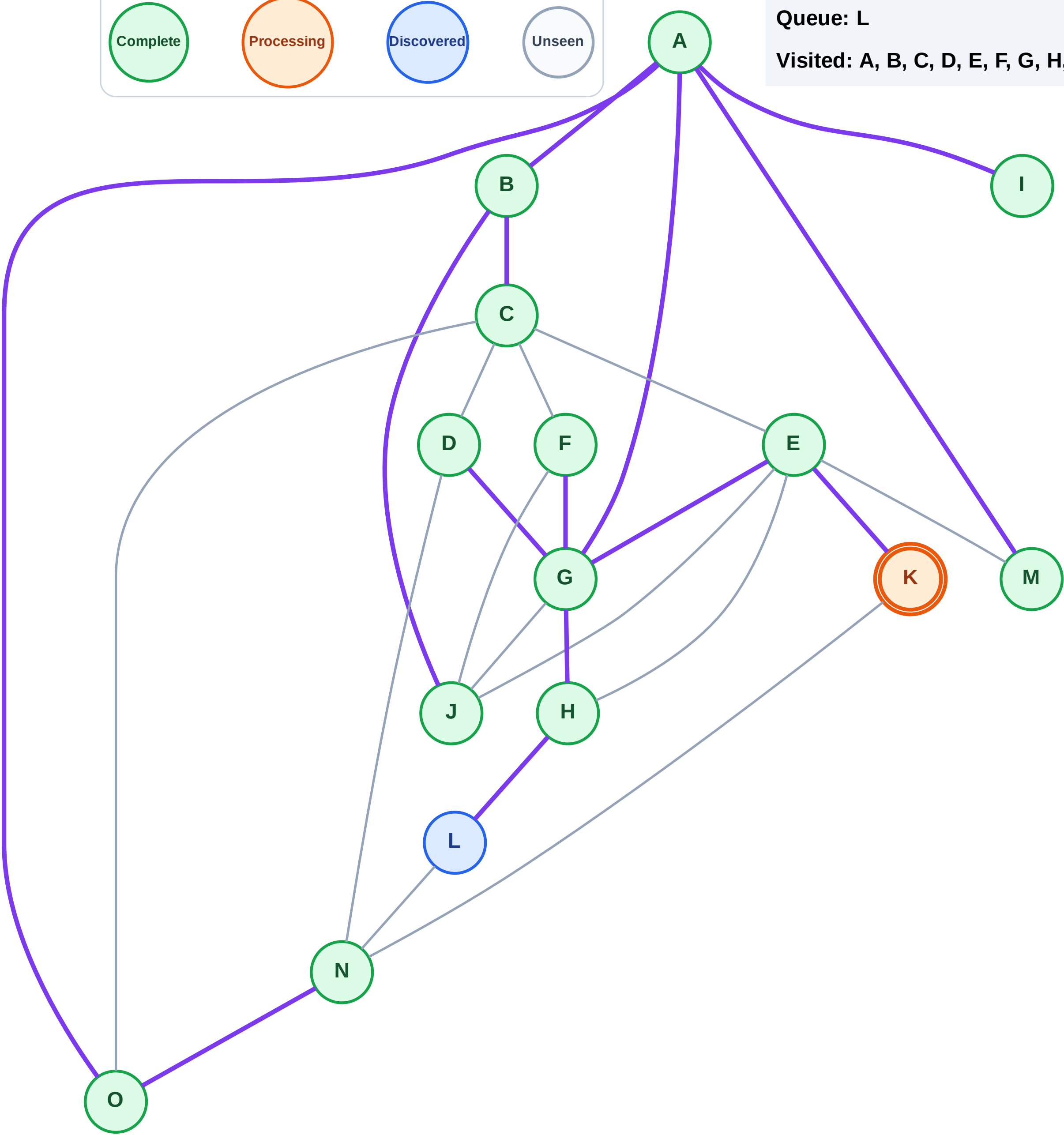
Processing

Discovered

Unseen

Queue: L

Visited: A, B, C, D, E, F, G, H, I, J, K, M, N, O



# DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

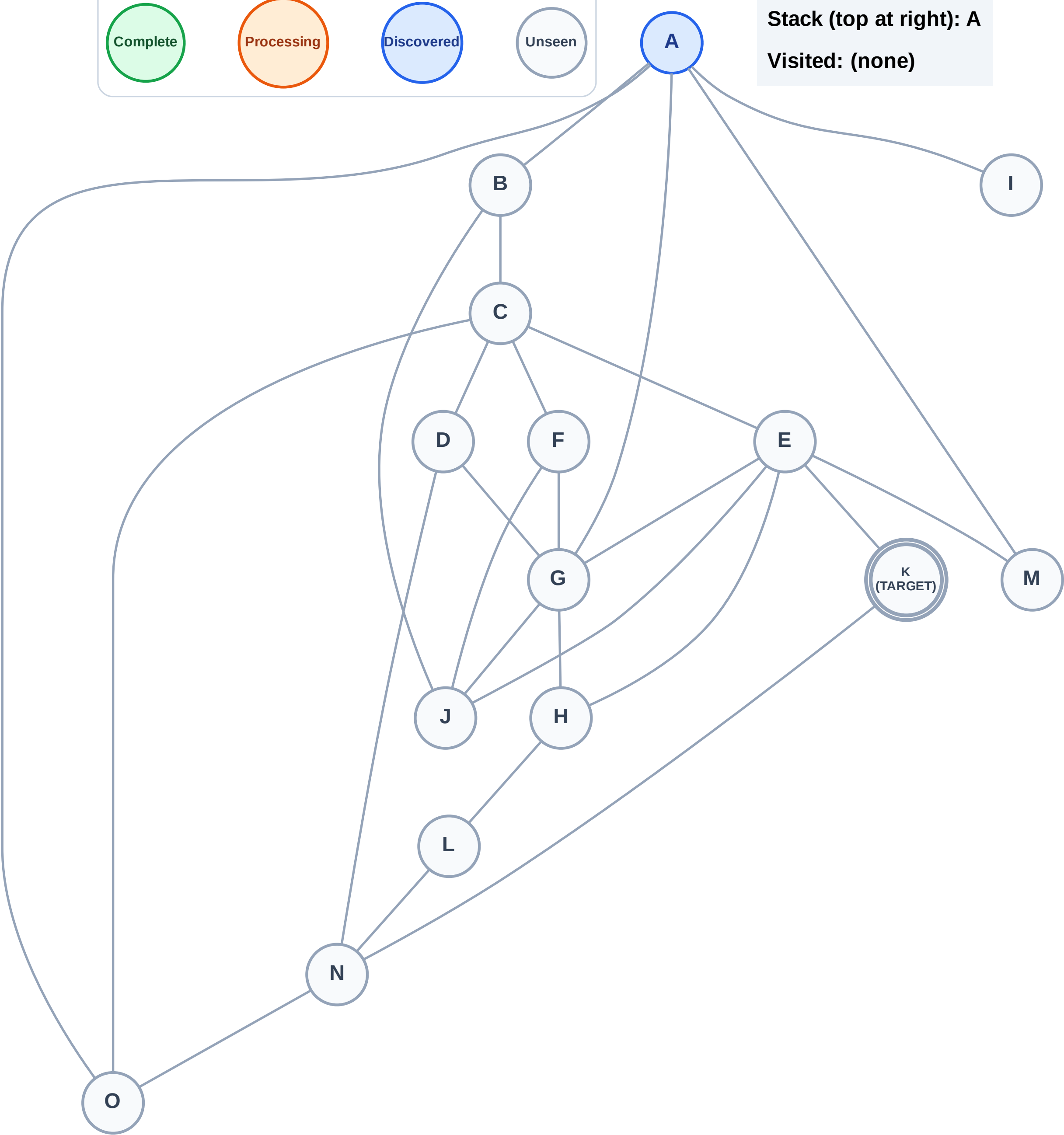
Complete

Processing

Discovered

Unseen

Stack (top at right): A  
Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

Complete

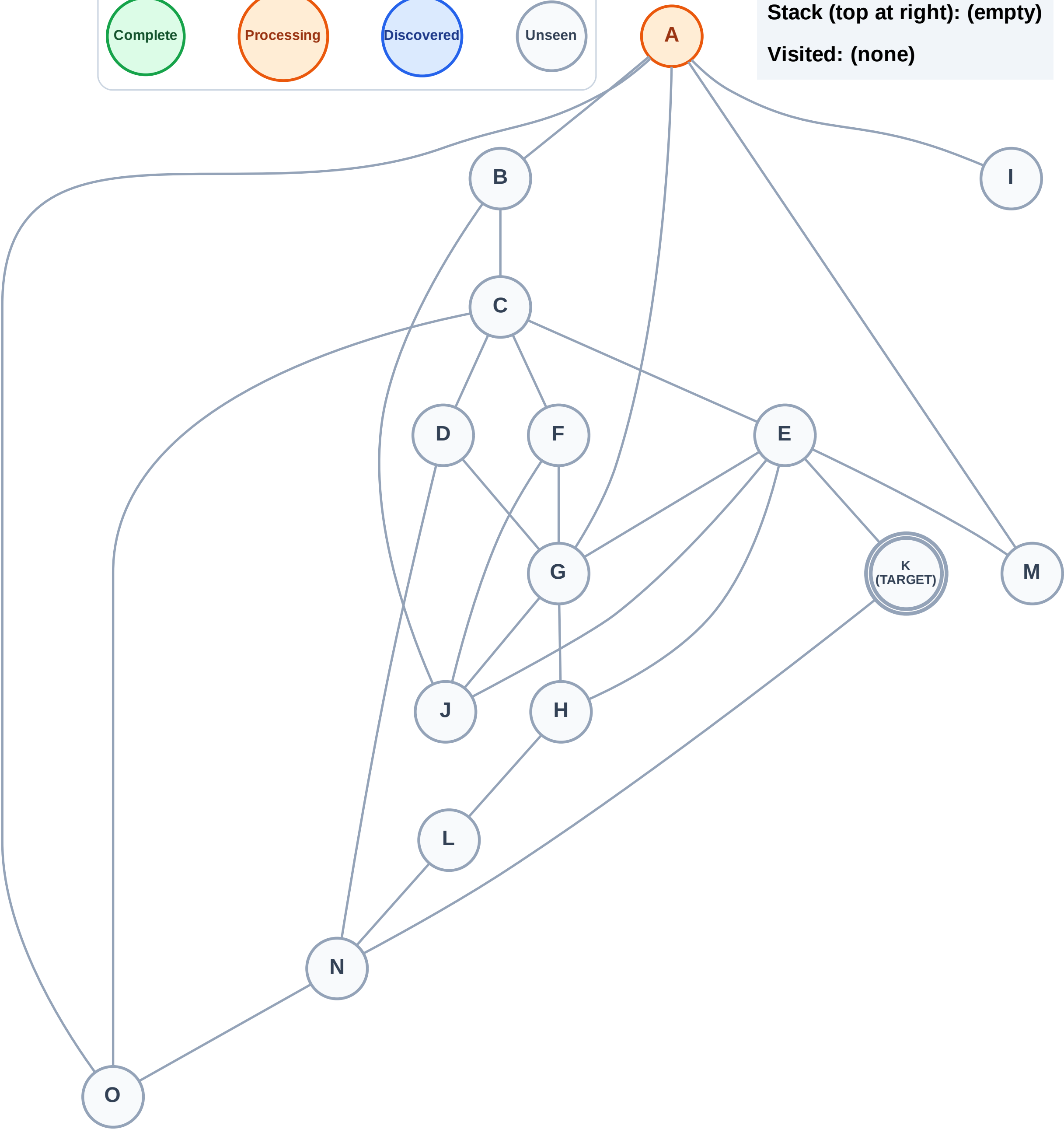
Processing

Discovered

Unseen

Stack (top at right): (empty)

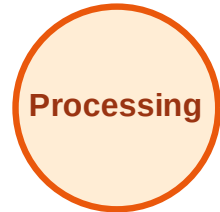
Visited: (none)



# DFS Traversal

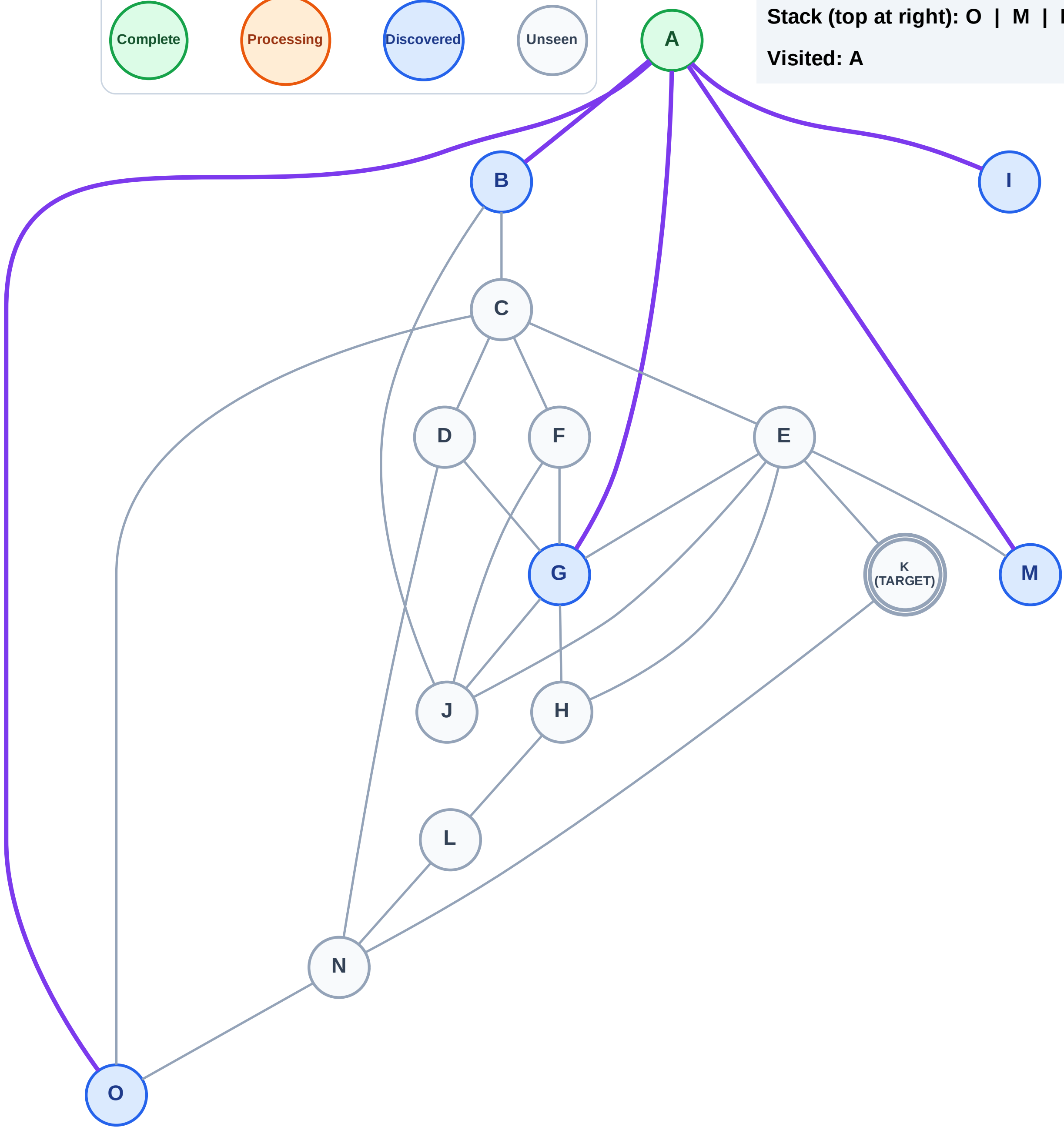
Step 3: Discover O, M, I, G, B from A

## Legend



Stack (top at right): O | M | I | G | B

Visited: A

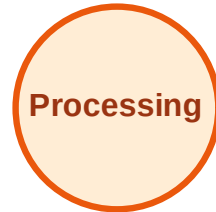




# DFS Traversal

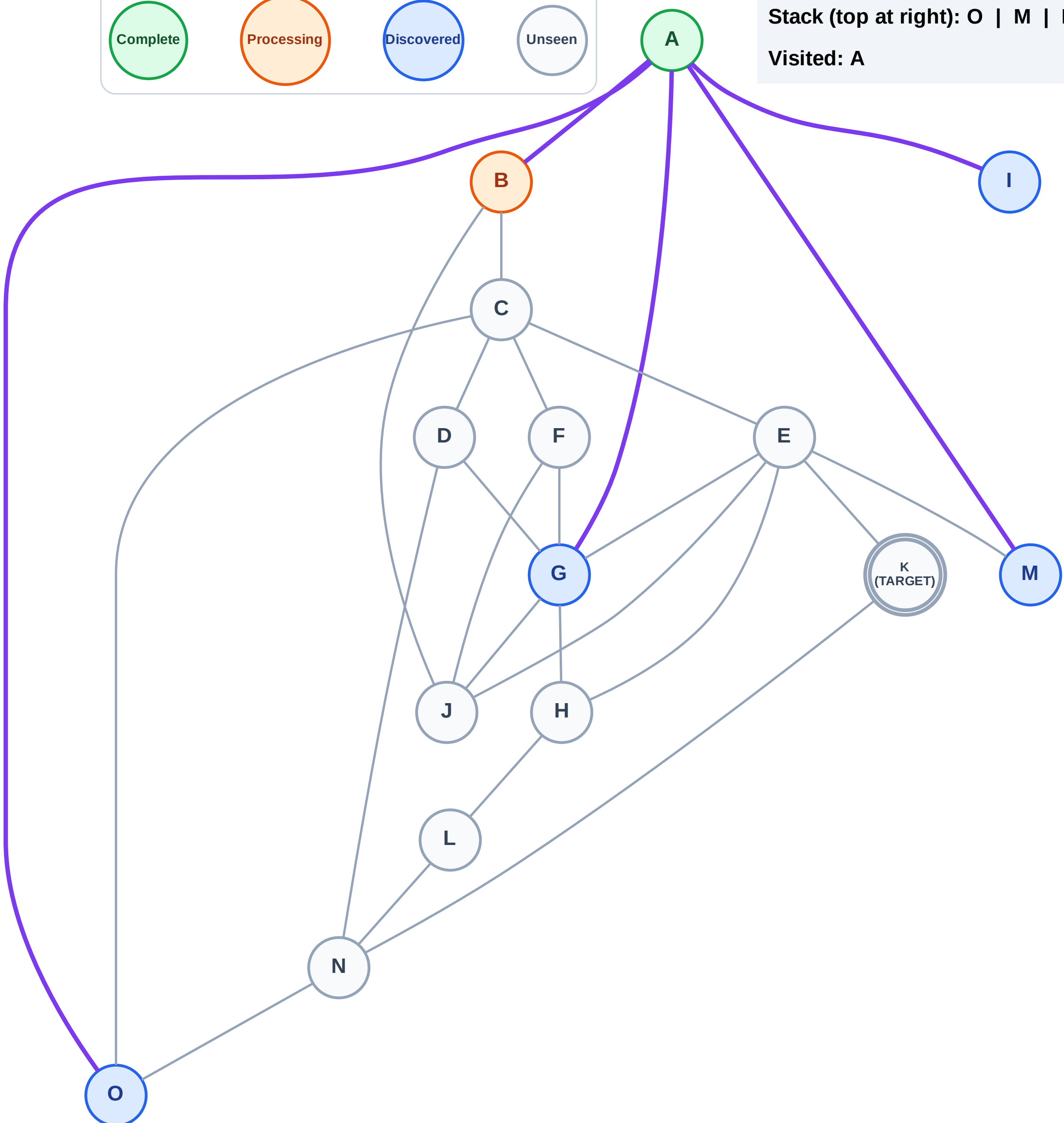
Step 4: Process node B

## Legend



Stack (top at right): O | M | I | G

Visited: A



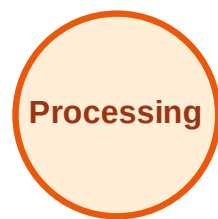
# DFS Traversal

Step 5: Discover J, C from B

## Legend



Complete



Processing



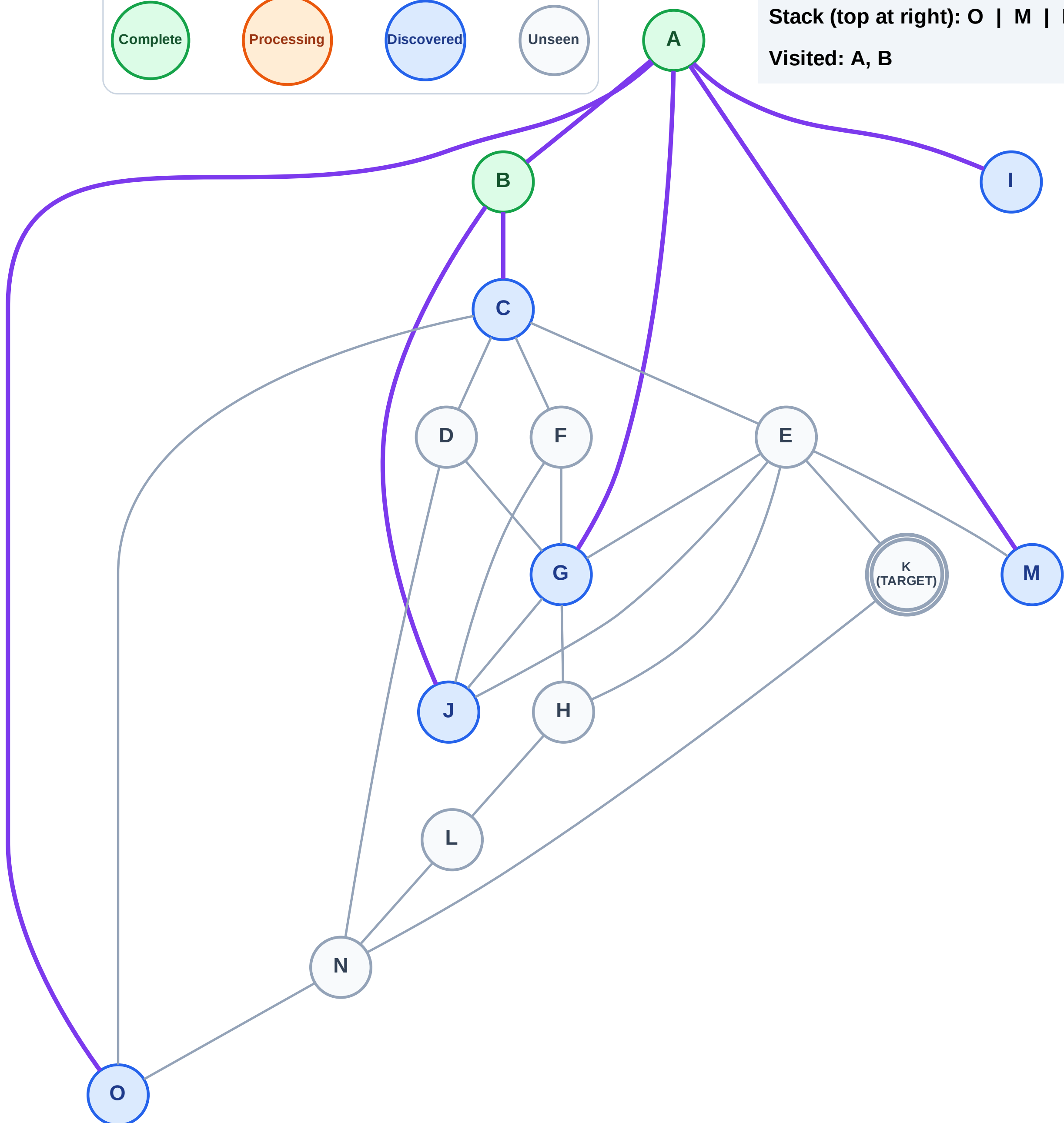
Discovered



Unseen

Stack (top at right): O | M | I | G | J | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

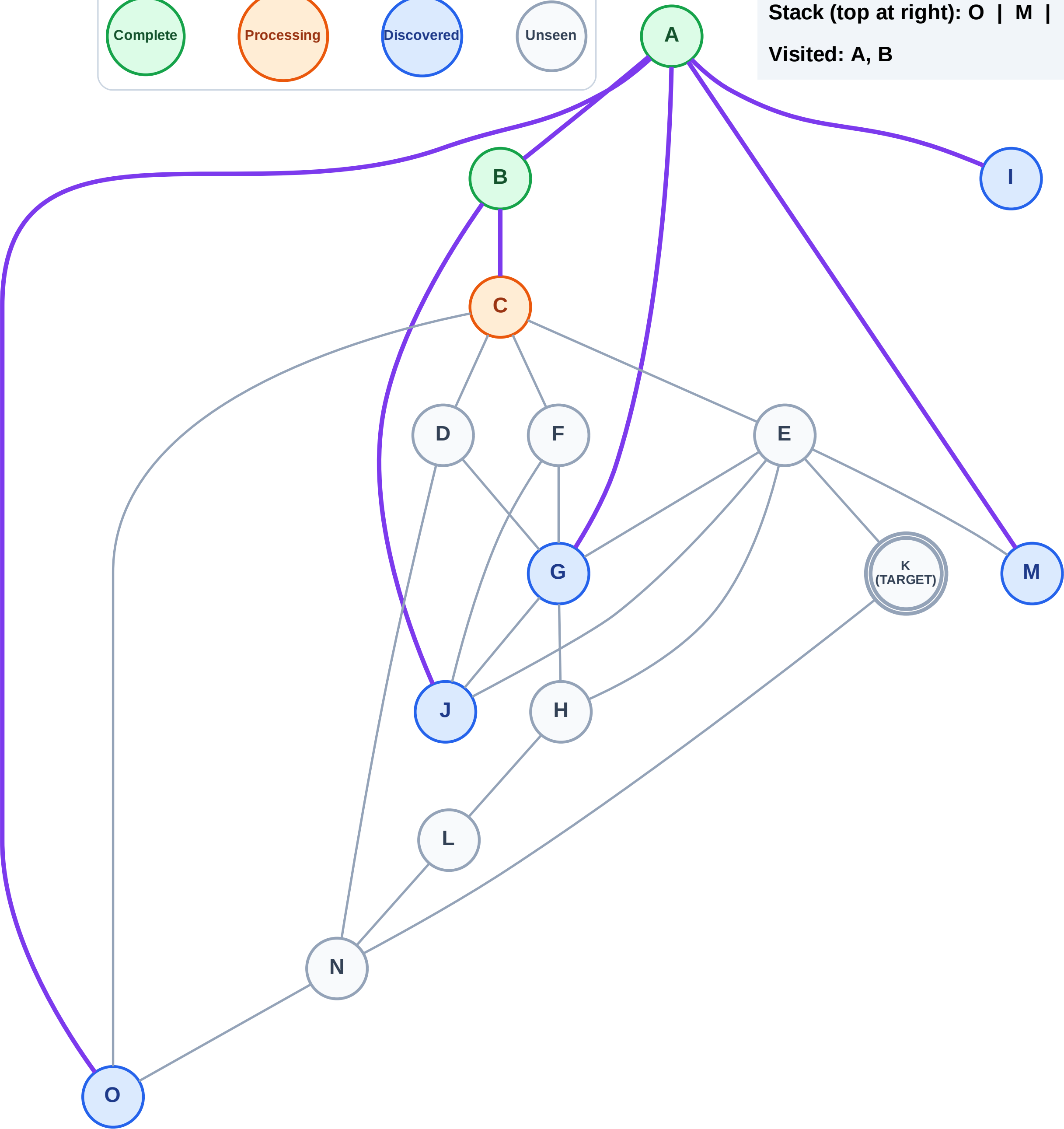
Processing

Discovered

Unseen

Stack (top at right): O | M | I | G | J

Visited: A, B



# DFS Traversal

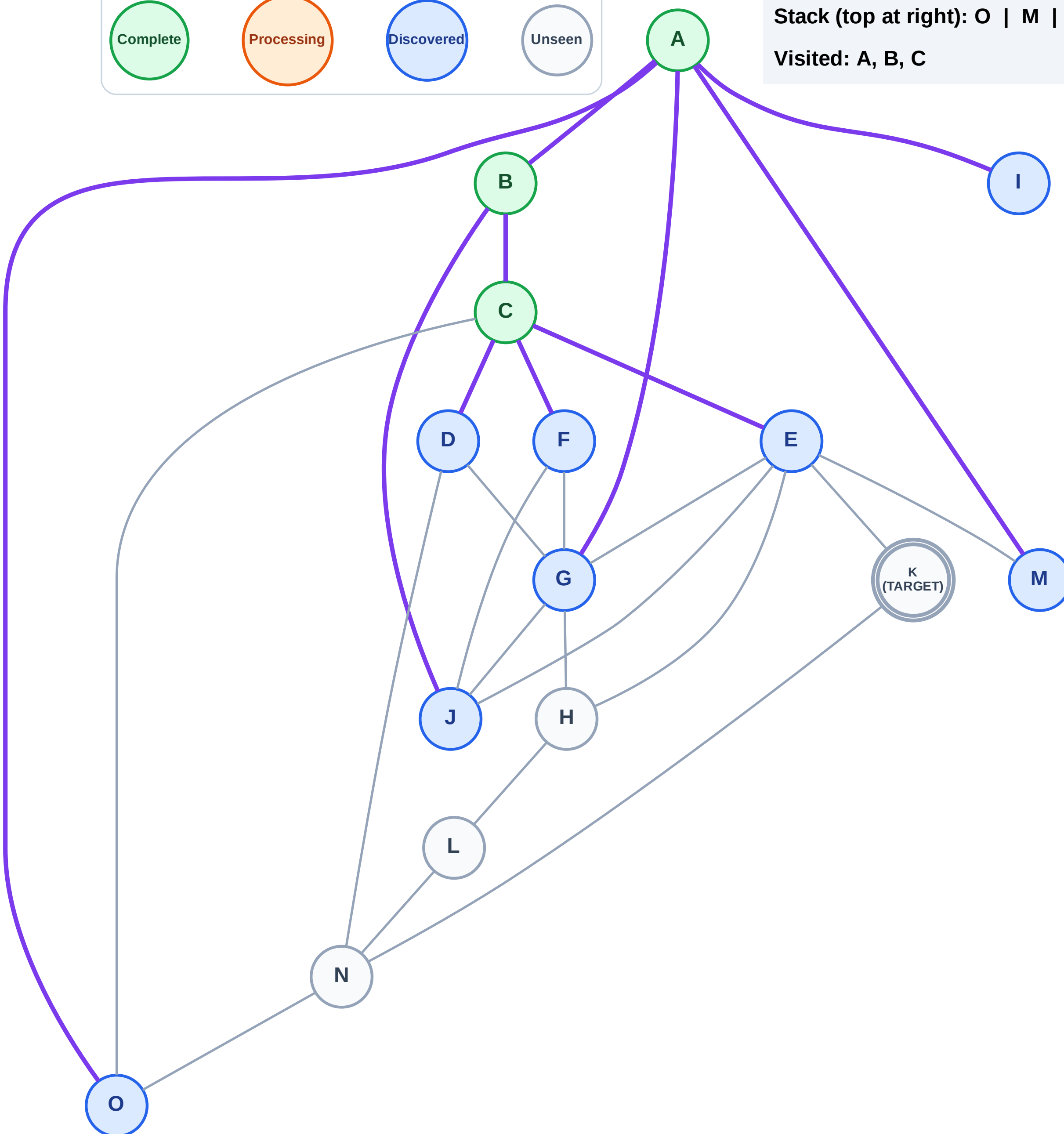
Step 7: Discover F, E, D from C

## Legend



Stack (top at right): O | M | I | G | J | F | E | D

Visited: A, B, C





# DFS Traversal

Step 8: Process node D

Legend

Complete

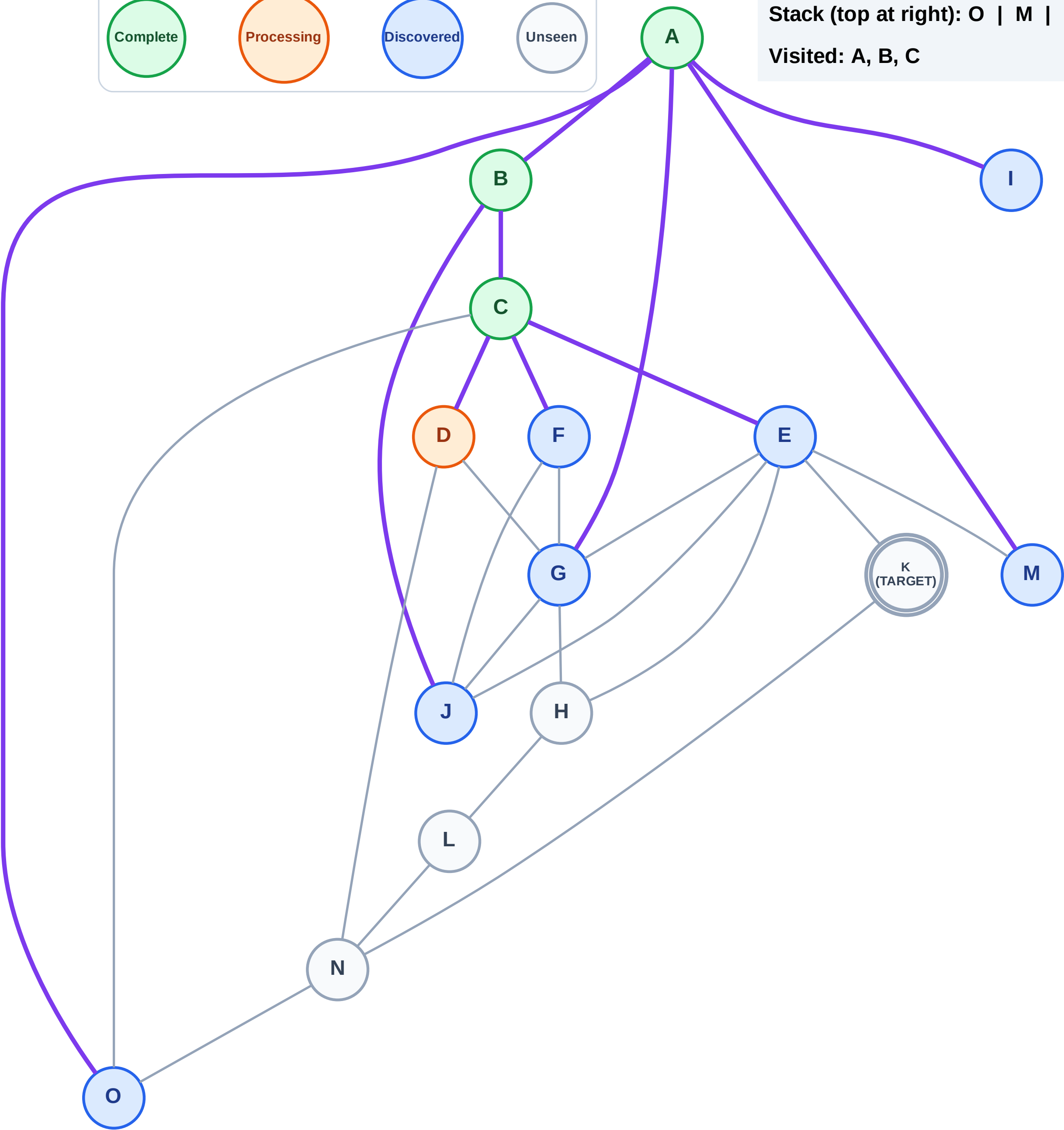
Processing

Discovered

Unseen

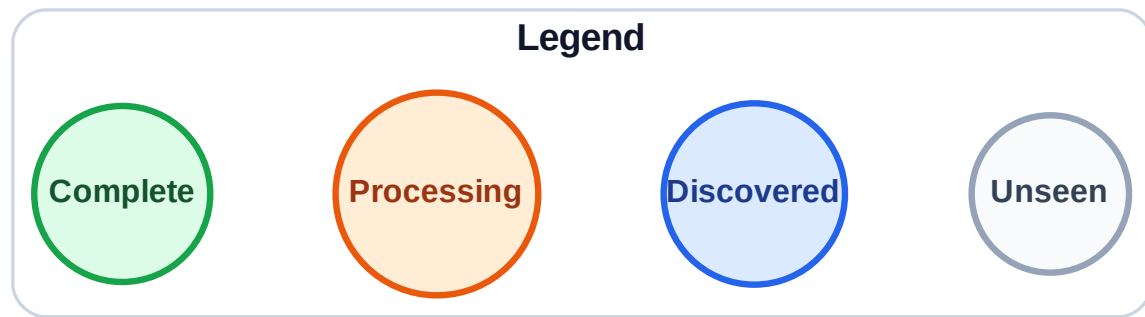
Stack (top at right): O | M | I | G | J | F | E

Visited: A, B, C



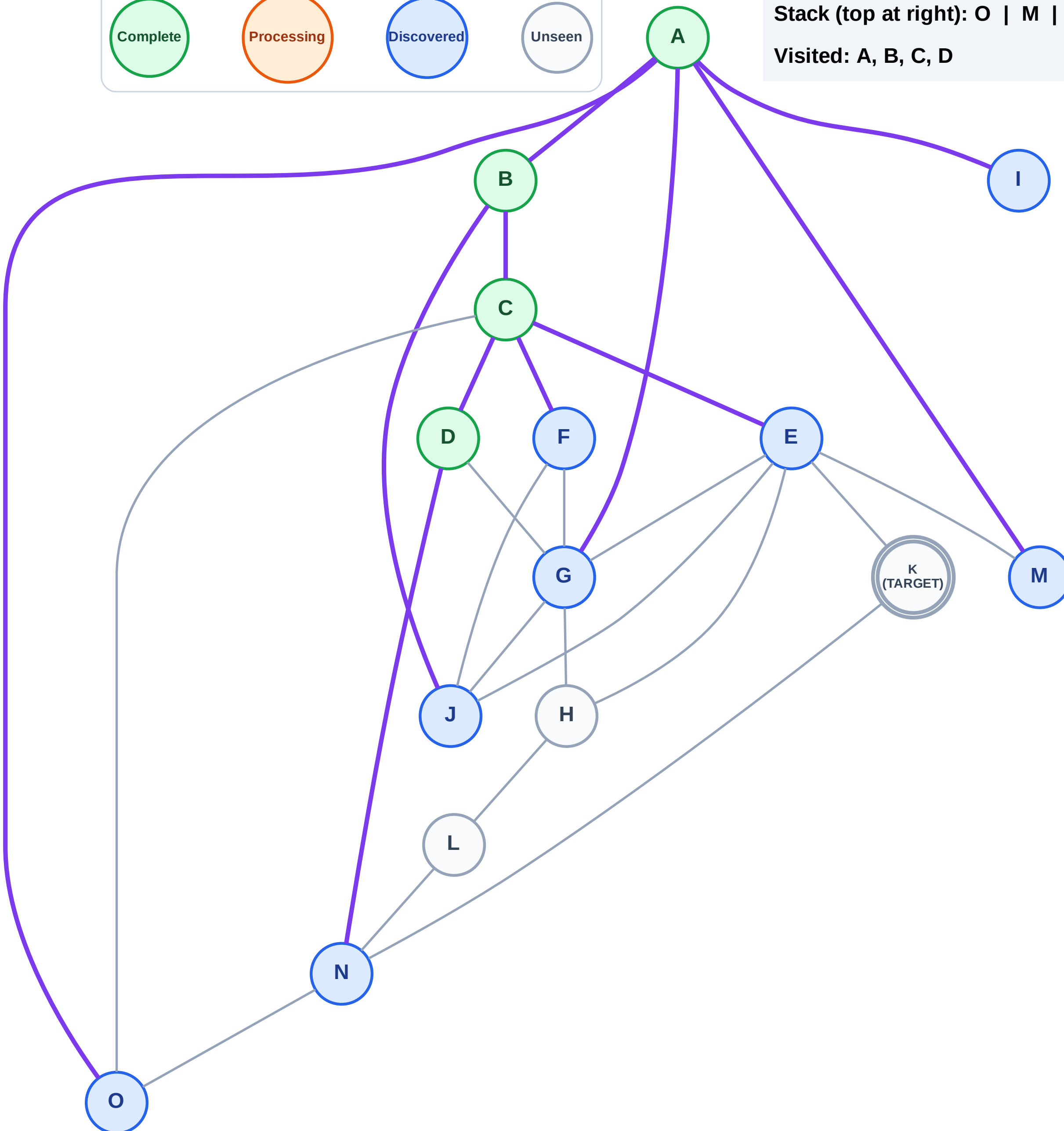
# DFS Traversal

Step 9: Discover N from D



Stack (top at right): O | M | I | G | J | F | E | N

Visited: A, B, C, D



# DFS Traversal

Step 10: Process node N

Legend

Complete

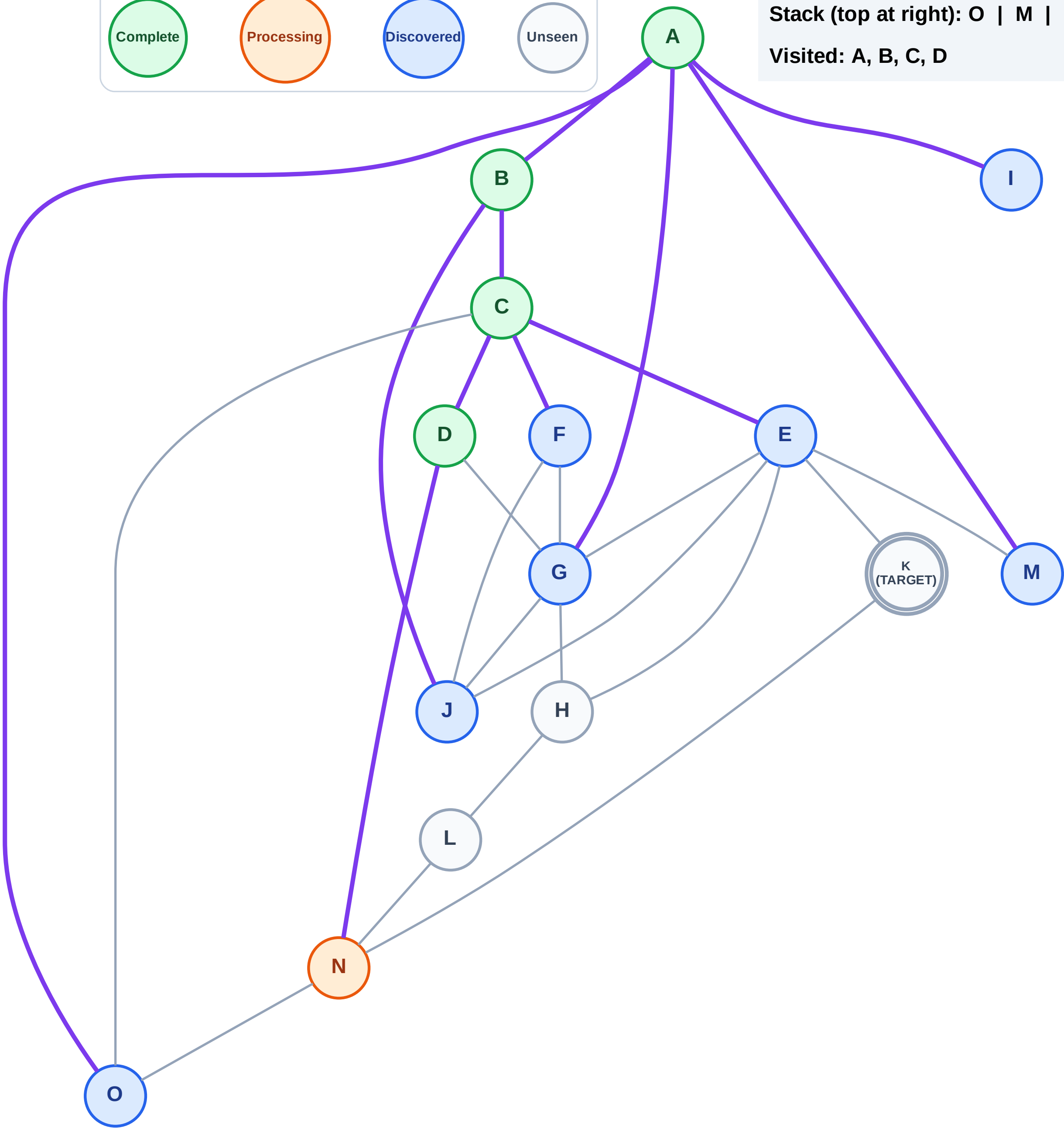
Processing

Discovered

Unseen

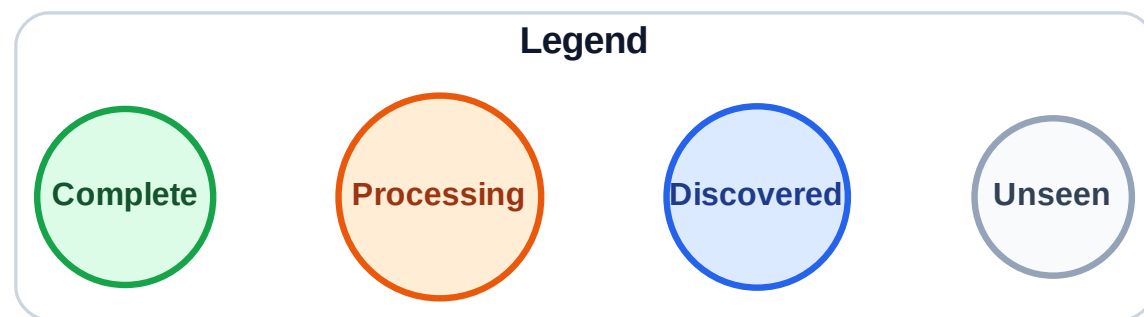
Stack (top at right): O | M | I | G | J | F | E

Visited: A, B, C, D

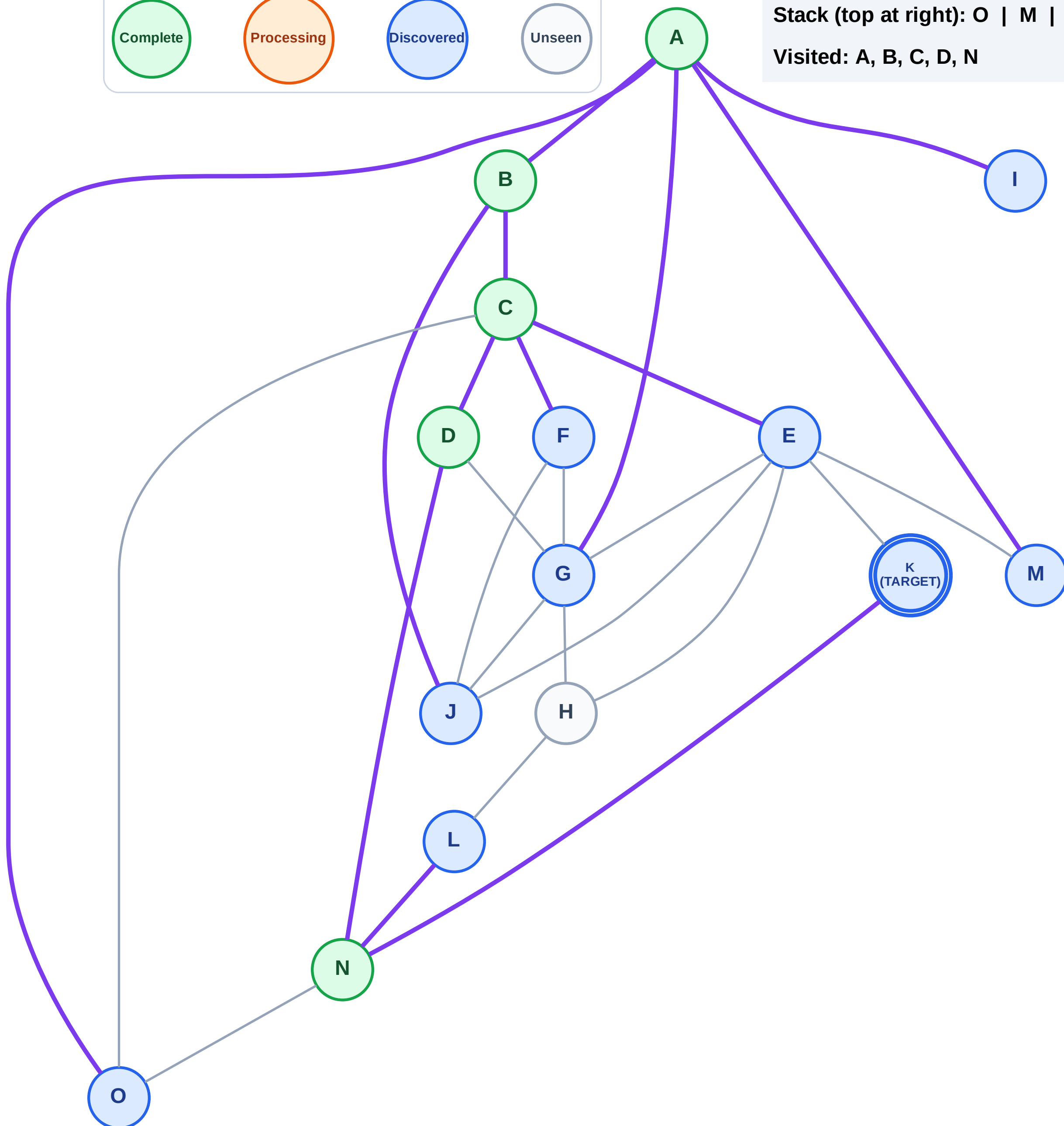


# DFS Traversal

Step 11: Discover L, K from N



Stack (top at right): O | M | I | G | J | F | E | L | K  
Visited: A, B, C, D, N





# DFS Traversal

Step 12: Process node K

Legend

Complete

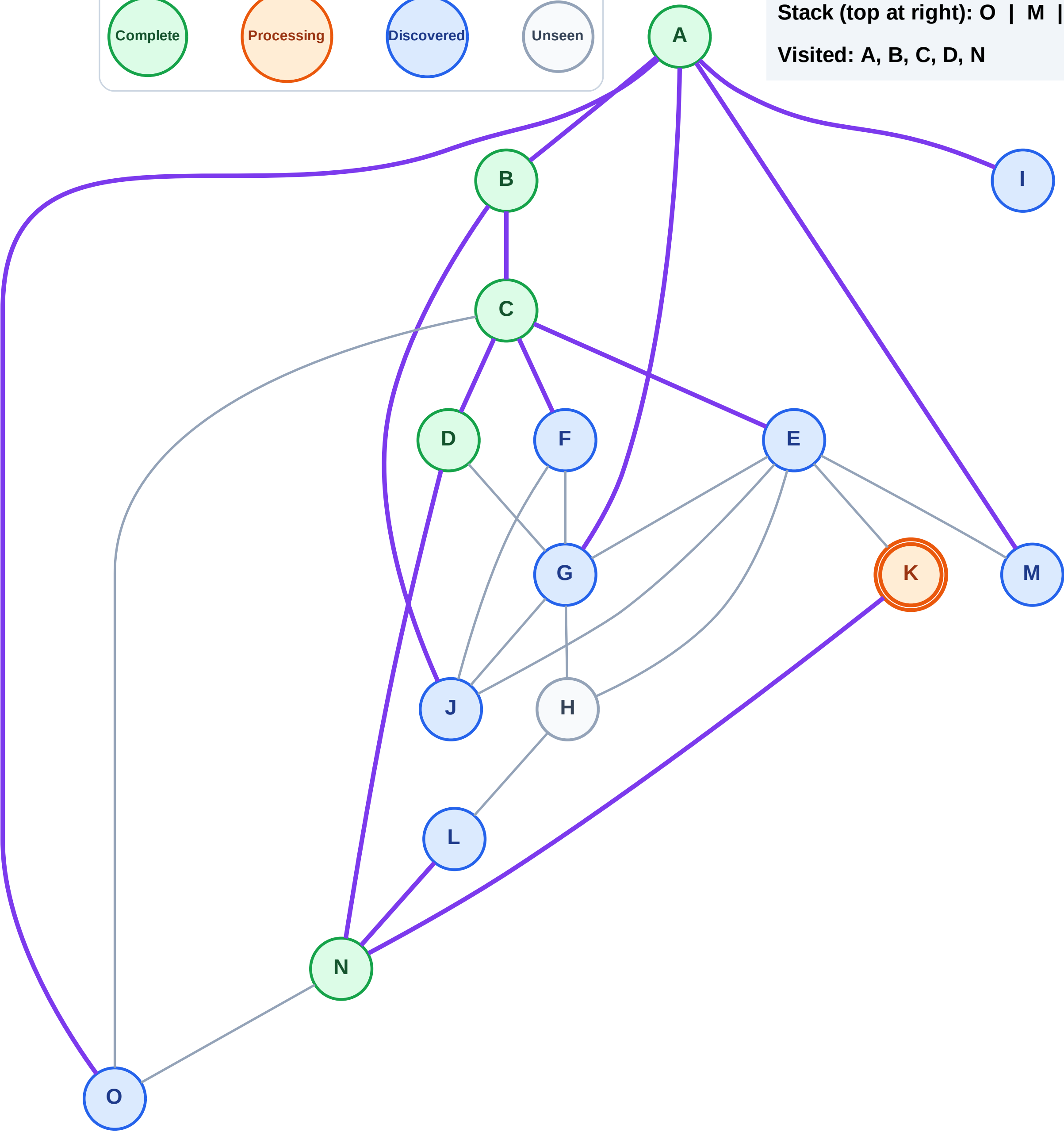
Processing

Discovered

Unseen

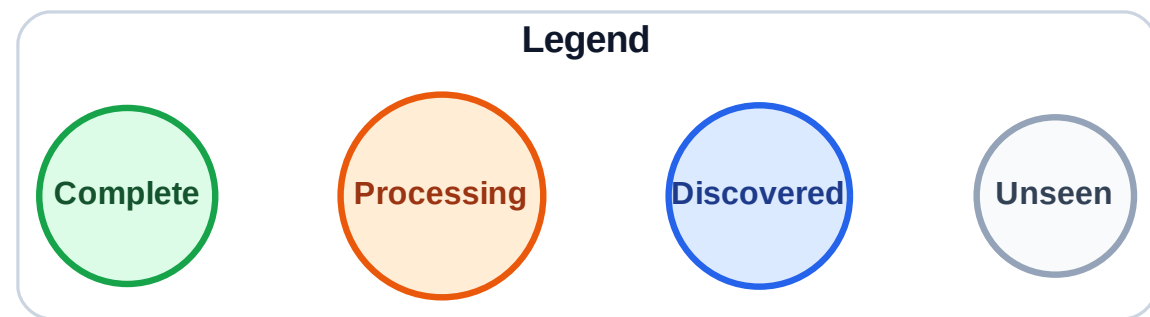
Stack (top at right): O | M | I | G | J | F | E | L

Visited: A, B, C, D, N



# DFS Traversal

Step 13: TARGET NODE K REACHED! Path (5 hops): A → B → C → D → N → K



**Stack (top at right):** O | M | I | G | J | F | E | L

**Visited: A, B, C, D, K, N**

